

Publications by Model: CARDAMOM

Total Publications: 97 (sorted newest to oldest)

1. Exploring Optimal Complexity for Water Stress Representation in Terrestrial Carbon Models: A Hybrid-Machine Learning Model Approach

Authors: J. Fang (Columbia University); P. Gentine (Columbia University)

Journal: Journal of Advances in Modeling Earth Systems

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2024ms004308

URL: <https://doi.org/10.1029/2024ms004308>

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References: 104

Cited by: 1 publications

Language: en

Subjects: N/A

Keywords: FLUXNET, model-data fusion, photosynthesis, water stress, evapotranspiration (and 1 more)

Funding: National Aeronautics and Space Administration, Schmidt Futures, European Research Council (and 1 more)

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Open Access: N/A

Abstract: Terrestrial biosphere models offer a comprehensive view of the global carbon cycle by integrating ecological processes across scales, yet they introduce significant uncertainties in climate and biogeochemical projections due to diverse process representations and parameter variations. For instance, different soil water limitation functions lead to wide productivity ranges across models. To address this, we propose the Differentiable Land Model (DifferLand), a novel hybrid machine learning approach replacing unknown water limitation functions in models with neural networks (NNs) to learn from data. Using automatic differentiation, we calibrated the embedded NN and the physical model parameters against daily observations of evapotranspiration, gross primary productivity, ecosystem respiration, and leaf area index across 16 FLUXNET sites. We evaluated six model configurations where NNs simulate increasingly complex soil water and photosynthesis interactions against test data sets to find the optimal structure-performance tradeoff. Our findings show that a simple hybrid model with a univariate NN effectively captures site-level water and carbon fluxes on a monthly timescale. Across a global aridity gradient, the magnitude of water stress limitation varies, but its functional form consistently converges to a piecewise linear relationship with saturation at high water levels. While models incorporating more interactions between soil water and meteorological drivers better fit observations at finer time scales, they risk overfitting and equifinality issues. Our study demonstrates

that hybrid models have great potential in learning unknown parameterizations and testing ecological hypotheses. Nevertheless, careful structure-performance tradeoffs are warranted in light of observational constraints to translate the retrieved relationships into robust process understanding.

2. Permafrost Region Greenhouse Gas Budgets Suggest a Weak CO₂ Sink and CH₄ and N₂O Sources, But Magnitudes Differ Between Top-Down and Bottom-Up Methods

Authors: G. Hugelius (Stockholm University); J. Ramage (Stockholm University); E. Burke (Met Office Hadley Centre, Met Office); A. Chatterjee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); T. L. Smallman (University of Edinburgh); T. Aalto (Finnish Meteorological Institute); A. Bastos (Max Planck Institute for Biogeochemistry); C. Biasi (University of Innsbruck, University of Eastern Finland); J. G. Canadell (Global Carbon Project CSIRO Environment Canberra ACT Australia); N. Chandra (Research Institute for Global Change JAMSTEC Yokohama Japan); F. Chevallier (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL (CEA-CNRS-UVSQ) Université Paris-Saclay 91191 Gif-sur-Yvette Gif-sur-Yvette France); P. Ciais (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL (CEA-CNRS-UVSQ) Université Paris-Saclay 91191 Gif-sur-Yvette Gif-sur-Yvette France); J. Chang (Zhejiang University); L. Feng (University of Edinburgh); M. W. Jones (University of East Anglia); T. Kleinen (Max Planck Institute for Meteorology); M. Kuhn (University of New Hampshire); R. Lauerwald (Université Paris-Saclay, INRAE, AgroParisTech, UMR Ecosys Palaiseau France); J. Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); E. López-Blanco (Greenland Institute of Natural Resources, Aarhus University); I. T. Lujckx (Wageningen University); M. E. Marushchak (University of Eastern Finland); S. M. Natali (Woodwell Climate Research Center Falmouth MA USA); Y. Niwa (Meteorological Research Institute, National Institute for Environmental Studies); D. Olefeldt (University of Alberta); P. I. Palmer (University of Edinburgh); P. K. Patra (Research Institute for Humanity and Research Kyoto Japan); W. Peters (Wageningen University, University of Groningen); S. Potter (Woodwell Climate Research Center Falmouth MA USA); B. Poulter (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); B. M. Rogers (Woodwell Climate Research Center Falmouth MA USA); W. J. Riley (Lawrence Berkeley National Laboratory); M. Saunois (Laboratoire des Sciences du Climat et de l'Environnement LSCE-IPSL (CEA-CNRS-UVSQ) Université Paris-Saclay 91191 Gif-sur-Yvette Gif-sur-Yvette France); E. A. G. Schuur (Northern Arizona University); R. L. Thompson (NILU Kjeller Norway); C. Treat (Alfred Wegener Institute); A. Tsuruta (Finnish Meteorological Institute); M. R. Turetsky (University of Colorado Boulder); A.-M. Virkkala (Woodwell Climate Research Center Falmouth MA USA); C. Voigt (University of Hamburg, University of Eastern Finland); J. Watts (Woodwell Climate Research Center Falmouth MA USA); Q. Zhu (Lawrence Berkeley National Laboratory); B. Zheng (Tsinghua University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gb007969

URL: <https://doi.org/10.1029/2023gb007969>

Publication Date: 2024-10-26

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References: 133

Cited by: N/A publications

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Abstract: Large stocks of soil carbon (C) and nitrogen (N) in northern permafrost soils are vulnerable to remobilization under climate change. However, there are large uncertainties in present-day greenhouse gas (GHG) budgets. We compare bottom-up (data-driven upscaling and process-based models) and top-down (atmospheric inversion models) budgets of carbon dioxide (CO₂), methane (CH₄) and nitrous oxide (N₂O) as well as lateral fluxes of C and N across the region over 2000-2020. Bottom-up approaches estimate higher land-to-atmosphere fluxes for all GHGs. Both bottom-up and top-down approaches show a sink of CO₂ in natural ecosystems (bottom-up: -29 (-709, 455), top-down: -587 (-862, -312) Tg CO₂-C yr⁻¹) and sources of CH₄ (bottom-up: 38 (22, 53), top-down: 15 (11, 18) Tg CH₄-C yr⁻¹) and N₂O (bottom-up: 0.7 (0.1, 1.3), top-down: 0.09 (-0.19, 0.37) Tg N₂O-N yr⁻¹). The combined global warming potential of all three gases (GWP-100) cannot be distinguished from neutral. Over shorter timescales (GWP-20), the region is a net GHG source because CH₄ dominates the total forcing. The net CO₂ sink in Boreal forests and wetlands is largely offset by fires and inland water CO₂ emissions as well as CH₄ emissions from wetlands and inland waters, with a smaller contribution from N₂O emissions. Priorities for future research include the representation of inland waters in process-based models and the compilation of process-model ensembles for CH₄ and N₂O. Discrepancies between bottom-up and top-down methods call for analyses of how prior flux ensembles impact inversion budgets, more and well-distributed in situ GHG measurements and improved resolution in upscaling techniques.

3. Greening of a boreal rich fen driven by CO₂ fertilisation

Authors: Sandeep Thayamkottu; T. Luke Smallman; Jaan Pärn; Ülo Mander; Eugénie S Euskirchen; Evan S Kane

Journal: Agricultural and Forest Meteorology

Publisher: Elsevier BV

DOI: 10.1016/j.agrformet.2024.110261

URL: <https://doi.org/10.1016/j.agrformet.2024.110261>

Publication Date: 2024-10-18

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References: 84

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

4. Carbon dioxide fertilization enhanced carbon sink offset by climate change and land use in Amazonia on a centennial scale

Authors: Bin Chen; Xiyan Xu; Shaoqiang Wang; Te Yang; Zhenhai Liu; Stefanie Falk

Journal: Science of The Total Environment

Publisher: Elsevier BV

DOI: 10.1016/j.scitotenv.2024.176903

URL: <https://doi.org/10.1016/j.scitotenv.2024.176903>

Publication Date: 2024-10-13

Volume/Issue: Volume 955

References: 71

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China

License: N/A

Open Access: N/A

Abstract: N/A

5. Estimating Global Gross Primary Production Using an Improved MODIS Leaf Area Index Dataset

Authors: Shujian Wang (Guangdong-Hong Kong Joint Laboratory for Carbon Neutrality, Jiangmen Laboratory of Carbon Science and Technology, Jiangmen 529199, China, Henan University); Xunhe Zhang (Henan University); Lili Hou (Guangdong-Hong Kong Joint Laboratory for Carbon Neutrality, Jiangmen Laboratory of Carbon Science and Technology, Jiangmen 529199, China, Henan University); Jiejie Sun (Guangdong-Hong Kong Joint Laboratory for Carbon Neutrality, Jiangmen Laboratory of Carbon Science and Technology, Jiangmen 529199, China); Ming Xu (Guangdong-Hong Kong Joint Laboratory for Carbon Neutrality, Jiangmen Laboratory of Carbon Science and Technology, Jiangmen 529199, China, Beijing Normal University, Henan University)

Journal: Remote Sensing

Publisher: MDPI AG

DOI: 10.3390/rs16193731

URL: <https://doi.org/10.3390/rs16193731>

Publication Date: 2024-10-08

Volume/Issue: Volume 16, Issue 19

References: 117

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Keywords: N/A

Funding: National Key Research and Development Program of China, Natural Science Foundation of Henan

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Open Access: N/A

Abstract: Remote sensing and process-coupled ecological models are widely used for the simulation of GPP, which plays a key role in estimating and monitoring terrestrial ecosystem productivity. However, most such models do not differentiate the C3 and C4 photosynthetic pathways and neglect the effect of nitrogen content on Vmax and Jmax, leading to considerable bias in the estimation of gross primary productivity (GPP). Here, we developed a model driven by the leaf area index, climate, and atmospheric CO2 concentration to estimate global GPP with a spatial resolution of 0.1 degrees and a temporal interval of 1 day from 2000 to 2022. We validated our model with ground-based GPP measurements at 128 flux tower sites, which yielded an accuracy of 72.3%. We found that the global GPP ranged from 116.4 PgCyear-1 to 133.94 PgCyear-1 from 2000 to 2022, with an average of 125.93 PgCyear-1. We also found that the global GPP showed an increasing trend of 0.548 PgCyear-1 during the study period. Further analyses using the structure equation model showed that atmospheric CO2 concentration and air temperature were the main drivers of the global GPP changes, total associations of 0.853 and 0.75, respectively, while precipitation represented a minor but negative contribution to global GPP.

6. Global Carbon Cycle Data Assimilation Using Earth Observation

Authors: Mathew Williams

Journal: Land Carbon Cycle Modeling

Publisher: CRC Press

DOI: 10.1201/9781032711126-34

URL: <https://doi.org/10.1201/9781032711126-34>

Publication Date: 2024-05-13

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References: N/A

Cited by: N/A publications

Language: en

Subjects: N/A

Keywords: Assimilation (phonology)|Earth observation

Funding: N/A

License: Creative Commons

Open Access: N/A

Abstract: N/A

7. Quantifying Regional Methane Emissions Using Airborne Transects and a Measurement-Model Fusion Approach

Authors: Andres Gonzalez (Universidad Mayor, University of Minnesota); Dylan B. Millet (University of Minnesota); Kelley C. Wells (University of Minnesota); Timothy J. Griffis (University of Minnesota); Zachary Barkley (Pennsylvania State University); Joshua P. DiGangi (National Aeronautics and Space Administration (NASA) Langley Research Center); John B. Nowak (National Aeronautics and Space Administration (NASA) Langley Research Center); Yonghoon Choi (National Aeronautics and Space Administration (NASA) Langley Research Center, Analytical Mechanics Associates)

Journal: ACS ES&T; Air

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acsestair.3c00072

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Volume/Issue: Volume 1, Issue 7

References: 72

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Language: en

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Keywords: N/A

Funding: Advanced Exploration Systems

License: N/A

Open Access: N/A

Abstract: N/A

8. Inferred drought-induced plant allocation shifts and their impact on drought legacy at a tropical forest site

Authors: Matthew A. Worden (Department of Earth System Science Stanford University Stanford California USA); Caroline A. Famiglietti (Department of Earth System Science Stanford University Stanford California USA, HydroSat Washington District of Columbia USA); Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Shuang Ma (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Damien Bonal (AgroParisTech, INRAE, UMR Silva Université de Lorraine Nancy France); Clément Stahl (INRAE, UMR EcoFoG, AgroParisTech, CNRS, Cirad Université des Antilles, Université de Guyane Kourou France); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford California USA)

Journal: Global Change Biology

Publisher: Wiley

DOI: 10.1111/gcb.17287

URL: <https://doi.org/10.1111/gcb.17287>

Publication Date: 2024-05-02

Volume/Issue: Volume 30, Issue 5

References: 99

Cited by: N/A publications

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Keywords: N/A

Funding: Agence Nationale de la Recherche, Division of Environmental Biology

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Open Access: N/A

Abstract: While droughts predominantly induce immediate reductions in plant carbon uptake, they can also exert long-lasting effects on carbon fluxes through associated changes in leaf area, soil carbon, etc. Among other mechanisms, shifts in carbon allocation due to water stress can contribute to the legacy effects of drought on carbon fluxes. However, the magnitude and impact of these allocation shifts on carbon fluxes and pools remain poorly understood. Using data from a wet tropical flux tower site in French Guiana, we demonstrate that drought-induced carbon allocation shifts can be reliably inferred by assimilating Net Biosphere Exchange (NBE) and other observations within the CARbon DATA MOdel fraMework. This model-data fusion system allows inference of optimized carbon and water cycle parameters and states from multiple observational data streams. We then examined how these inferred shifts affected the duration and magnitude of drought's impact on NBE during and after the extreme event. Compared to a static allocation scheme analogous to those typically implemented in land surface models, dynamic allocation reduced average carbon uptake during drought recovery by a factor of 2.8. Additionally, the dynamic model extended the average recovery time by 5 months. The inferred allocation shifts influenced the post-drought period by altering foliage and fine root pools, which in turn modulated gross primary productivity and heterotrophic respiration for up to a decade. These changes can create a bust-boom cycle where carbon uptake is enhanced some years after a drought,

compared to what would have occurred under drought-free conditions. Overall, allocation shifts accounted for 65% [45%-75%] of drought legacy effects in modeled NBE. In summary, drought-induced carbon allocation shifts can play a substantial role in the enduring influence of drought on cumulative land-atmosphere CO₂ exchanges and should be accounted for in ecosystem models.

9. Satellite-detected large CO₂ release in southwestern North America during the 2020-2021 drought and associated wildfires

Authors: Hui Chen; Wei He; Jinxiu Liu; Ngoc Tu Nguyen; Frédéric Chevallier; Hua Yang; Yiming Lv; Chengcheng Huang; Christian Rödenbeck; Scot M Miller; Fei Jiang; Junjie Liu; Matthew S Johnson; Sajeev Philip; Zhiqiang Liu; Ning Zeng; Sourish Basu; David F Baker

Journal: Environmental Research Letters

Publisher: IOP Publishing

DOI: 10.1088/1748-9326/ad3cf7

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Publication Date: 2024-04-10

Volume/Issue: Volume 19, Issue 5

References: 81

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Language: N/A

Subjects: N/A

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Open Access: N/A

Abstract: Southwestern North America (SWNA) continuously experienced megadroughts and large wildfires in 2020 and 2021. Here, we quantified their impact on the terrestrial carbon budget using net biome production (NBP) estimates from an ensemble of atmospheric inversions assimilating in-situ CO₂ and Carbon Observatory-2 (OCO-2) satellite XCO₂ retrievals (OCO-2 v10 MIP Extension), two satellite-based gross primary production (GPP) datasets, and two fire CO₂ emission datasets. We found that the 2020-2021 drought and associated wildfires in SWNA led to a large CO₂ loss, an ensemble mean of 95.07 TgC estimated by the satellite inversions using both nadir and glint XCO₂ retrievals (LNLG) within the OCO-2 v10 MIP, greater than 80% of SWNA's annual total carbon sink. Moreover, the carbon loss in 2020 was mainly contributed by fire emissions while in 2021 mainly contributed by drought impacts on terrestrial carbon uptake. In addition, the satellite inversions indicated the huge carbon loss was mainly contributed by fire emissions from forests and grasslands along with carbon uptake reductions due to drought impacts on grasslands and shrublands. This study provides a process understanding of how some droughts and following wildfires affect the terrestrial carbon budget on a regional scale.

10. Underestimated Dry Season Methane Emissions from Wetlands in the Pantanal

Authors: Mengze Li (Department of Earth System Science, Stanford University, Stanford, California 94305, United States, University of Michigan); Eric A. Kort (University of Michigan); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Dien Wu (California Institute of Technology); Genevieve Plant (University of Michigan); Cynthia Gerlein-Safdi (University of California Berkeley); Tianjiao Pu (University of California Berkeley)

Journal: Environmental Science & Technology

Publisher: American Chemical Society (ACS)

DOI: 10.1021/acs.est.3c09250

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References: 42

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Keywords: N/A

Funding: National Aeronautics and Space Administration

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Open Access: N/A

Abstract: N/A

11. Widespread temporal and spatial variability in net ecosystem productivity under climate change

Authors: Heng Huang; Ignacio Rodriguez-Iturbe; Salvatore Calabrese

Journal: One Earth

Publisher: Elsevier BV

DOI: 10.1016/j.oneear.2024.01.008

URL: <https://doi.org/10.1016/j.oneear.2024.01.008>

Publication Date: 2024-02-02

Volume/Issue: Volume 7, Issue 3

References: 64

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

12. Spatial biases reduce the ability of Earth system models to simulate soil heterotrophic respiration fluxes

Authors: Bertrand Guenet; Jérémie Orliac; Lauric Cécillon; Olivier Torres; Laura Sereni; Philip A. Martin; Pierre Barré; Laurent Bopp

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-21-657-2024

URL: <https://doi.org/10.5194/bg-21-657-2024>

Publication Date: 2024-02-02

Volume/Issue: Volume 21, Issue 2

References: 52

Cited by: 4 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: . Heterotrophic respiration (Rh) is, at a global scale, one of the largest CO₂ fluxes between the Earth's surface and atmosphere and may increase in the future. The previous generation of Earth system models (ESMs) was able to reproduce global fluxes relatively well, but at that, time no gridded products were available to perform an in-depth evaluation. The capacity of the new generation of ESMs used within the Coupled Model Intercomparison Project Phase 6 (CMIP6) to reproduce this flux has not been evaluated, meaning that the realism of resulting CO₂ flux estimates is unclear. In this study, we combine recently released observational data on Rh and ESM simulations to evaluate the ability of 13 ESMs from CMIP6 to reproduce Rh. Only 4 of the 13 tested ESMs were able to reproduce the total Rh flux, but spatial analysis underlined important bias compensation for most of the ESMs, which generally showed an overestimation in tropical regions and an underestimation in arid regions. To identify the main drivers of the bias, we performed an analysis of the residuals and found that mean annual precipitation was the most important driver explaining the difference between ESM simulations and observation-derived products of Rh, with a higher bias between ESM simulations and Rh products where precipitation was high. Based on our results, next-generation ESMs should focus on improving the response of Rh to soil moisture.

13. Synthesis of the land carbon fluxes of the Amazon region between 2010 and 2020

Authors: Thais M. Rosan; Stephen Sitch; Michael O'Sullivan; Luana S. Basso; Chris Wilson; Camila Silva; Emanuel Gloor; Dominic Fawcett; Viola Heinrich; Jefferson G. Souza; Francisco Gilney Silva Bezerra; Celso von Randow; Lina M. Mercado; Luciana Gatti; Andy Wiltshire; Pierre Friedlingstein; Julia Pongratz; Clemens Schwingshackl; Mathew Williams; Luke Smallman; Jürgen Knauer; Vivek Arora; Daniel Kennedy; Hanqin Tian; Wenping Yuan; Atul K. Jain; Stefanie Falk; Benjamin Poulter; Almut Arneth; Qing Sun; Sönke Zaehle; Anthony P. Walker; Etsushi Kato; Xu Yue; Ana Bastos; Philippe Ciais; Jean-Pierre Wigneron; Clement Albergel; Luiz E. O. C. Aragão

Journal: Communications Earth & Environment

Publisher: Springer Science and Business Media LLC

DOI: 10.1038/s43247-024-01205-0

URL: <https://doi.org/10.1038/s43247-024-01205-0>

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References: 83

Cited by: 9 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: The Amazon is the largest continuous tropical forest in the world and plays a key role in the global carbon cycle. Human-induced disturbances and climate change have impacted the Amazon carbon balance. Here we conduct a comprehensive synthesis of existing state-of-the-art estimates of the contemporary land carbon fluxes in the Amazon using a set of bottom-up methods (i.e., dynamic vegetation models and bookkeeping models) and a top-down inversion (atmospheric inversion model) over the Brazilian Amazon and the whole Biogeographical Amazon domain. Over the whole biogeographical Amazon region bottom-up methodologies suggest a small average carbon sink over 2010-2020, in contrast to a small carbon source simulated by top-down inversion (2010-2018). However, these estimates are not significantly different from one another when accounting for their large individual uncertainties, highlighting remaining knowledge gaps, and the urgent need to reduce such uncertainties. Nevertheless, both methodologies agreed that the Brazilian Amazon has been a net carbon source during recent climate extremes and that the south-eastern Amazon was a net land carbon source over the whole study period (2010-2020). Overall, our results point to increasing human-induced disturbances (deforestation and forest degradation by wildfires) and reduction in the old-growth forest sink during drought.

14. Water Stress Dominates 21st-Century Tropical Land Carbon Uptake

Authors: Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Victoria Meyer (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA); Marcos Longo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, Lawrence Berkeley National Laboratory)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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References: 94

Cited by: 4 publications

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Subjects: N/A

Keywords: data assimilation, carbon cycle, water stress

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Open Access: N/A

Abstract: Water stress regulates land-atmosphere carbon dioxide (CO₂) exchanges in the tropics; however, its role remains poorly characterized due to the confounding roles of radiation, temperature and canopy dynamics. In particular, uncertainty stems from the relative roles of plant-available water (supply) and atmospheric water vapor deficit (demand) as mechanistic drivers of photosynthetic carbon (C) uptake variability. Using satellite measurements of gravity, CO₂ and fluorescence to constrain a mechanistic carbon-water cycle model from 2001 to 2018, we found that the interannual variability (IAV) of water stress on photosynthetic C uptake was 52% greater than the combined effects of other factors. Surprisingly, the dominance of water stress on C uptake IAV was greater in the wet tropics (94%) than in the dry tropics (26%). Plant-available water supply and atmospheric demand both contributed to the IAV of water stress on photosynthetic C uptake across the tropics, but the IAV of demand effects was 21% greater than the IAV of supply effects (33% greater in the wet tropics and 6% greater in the dry tropics). We found that the IAV of water stress on C uptake was 24% greater than the IAV of the combination of other factors in the net land-atmosphere C sink in the whole tropics, 26% greater in the wet tropics, and 7% greater in the dry tropics. Given the recent trends in tropical

precipitation and atmospheric humidity, our findings indicate that water stress----from both supply and demand----will likely dominate the climate response of land C sink across tropical ecosystems in the coming decades.

15. Emergent Climatic Controls on Soil Carbon Turnover and Its Variability in Warm Climates

Authors: Heng Huang (Department of Biological and Agricultural Engineering Texas A&M; University College Station TX USA, The University of Hong Kong); Jin Wu (The University of Hong Kong); Salvatore Calabrese (Department of Biological and Agricultural Engineering Texas A&M; University College Station TX USA)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2023gl105291

URL: <https://doi.org/10.1029/2023gl105291>

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Cited by: 2 publications

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Subjects: N/A

Keywords: N/A

Funding: National Science Foundation, National Institute of Food and Agriculture

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Open Access: N/A

Abstract: Climate plays a critical role in altering soil carbon (C) turnover and long-term soil C storage by regulating water availability and temperature, and in turn biological activity. However, a systematic analysis of how key climatic factors shape the global patterns of soil C turnover is still lacking. Using global observation-based data sets and a transit time theory, here we show that--excluding croplands and cold regions--soil C turnover time (τ_{TO}) and its variability are strongly related to ecosystem aridity through a power law scaling. According to such a relation, soil C turnover is faster but also more variable in wetter regions, suggesting more complex C cycling processes. The observed scaling of τ_{TO} and its coefficient of variation with aridity underlines the fundamental controls of climate on soil C turnover and may help reconcile soil C models with empirical observations for improved projection of soil C dynamics under climate change.

16. RECCAP2 Future Component: Consistency and Potential for Regional Assessment to Constrain Global Projections

Authors: Chris D. Jones (Met Office Hadley Centre, Met Office); Tilo Ziehn (CSIRO Environment Aspendale VIC Australia); Jatin Anand (University of Illinois); Ana Bastos (Department Biogeochemical Integratio Max-Planck-Institut für Biogeochemie Jena Germany); Eleanor Burke (Met Office Hadley Centre, Met Office); Josep G. Canadell (Global Carbon Project CSIRO Environment Canberra ACT Australia); Manoel Cardoso (National Institute for Space Research São José dos Campos Brazil); Yolandi Ernst (University of the Witwatersrand); Atul K. Jain (University of Illinois); Sujong Jeong (Seoul National University); Elizabeth D. Keller (GNS Science Lower Hutt New Zealand, Victoria University of Wellington); Masayuki Kondo (Nagoya University); Ronny Lauerwald (INRAE/AgroParisTech-UMR Ecosys Université Paris-Saclay Palaiseau France); Tzu-Shun Lin (University of Illinois, National Center for Atmospheric Research); Guillermo Murray-Tortarolo (Instituto de Investigaciones en Ecosistemas y Sustentabilidad Universidad Nacional Autónoma de México Morelia México); Gert-Jan Nabuurs (Wageningen Environmental Research Wageningen The Netherlands); Mike O'Sullivan (University of Exeter); Ben Poulter (National Aeronautics and Space Administration (NASA) Goddard Space Flight Center); Xiaoyu Qin (Chinese Academy of Sciences); Celso von Randow (National Institute for Space Research São José dos Campos Brazil); Marcos Sanches (National Institute for Space Research São José dos Campos Brazil); Dmitry Schepaschenko (International Institute for Applied Systems Analysis Laxenburg Austria); Anatoly Shvidenko (International Institute for Applied Systems Analysis Laxenburg Austria); T. Luke Smallman (University of Edinburgh); Hanqin Tian (Department of Earth and Environmental Sciences Schiller Institute for Integrated Science and Society Boston College Chestnut Hill MA USA); Yohanna Villalobos (Global Carbon Project CSIRO Environment Canberra ACT Australia); Xuhui Wang (Peking University); Jeongmin Yun (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: Projections of future carbon sinks and stocks are important because they show how the world's ecosystems will respond to elevated CO₂ and changes in climate. Moreover, they are crucial to inform policy decisions around emissions reductions to stay within the global warming levels identified by the Paris Agreement. However, Earth System Models from the 6th Coupled Model Intercomparison Project (CMIP6) show substantial spread in future projections--especially of the terrestrial carbon cycle,

leading to a large uncertainty in our knowledge of any remaining carbon budget (RCB). Here we evaluate the global terrestrial carbon cycle projections on a region-by-region basis and compare the global models with regional assessments made by the REgional Carbon Cycle Assessment and Processes, Phase 2 activity. Results show that for each region, the CMIP6 multi-model mean is generally consistent with the regional assessment, but substantial cross-model spread exists. Nonetheless, all models perform well in some regions and no region is without some well performing models. This gives confidence that the CMIP6 models can be used to look at future changes in carbon stocks on a regional basis with appropriate model assessment and benchmarking. We find that most regions of the world remain cumulative net sources of CO₂ between now and 2100 when considering the balance of fossil-fuels and natural sinks, even under aggressive mitigation scenarios. This paper identifies strengths and weaknesses for each model in terms of its performance over a particular region including how process representation might impact those results and sets the agenda for applying stricter constraints at regional scales to reduce the uncertainty in global projections.

17. Modeling China's terrestrial ecosystem gross primary productivity with BEPS model: Parameter sensitivity analysis and model calibration

Authors: Xiuli Xing; Mousong Wu; Wenxin Zhang; Weimin Ju; Torbern Tagesson; Wei He; Songhan Wang; Jun Wang; Lu Hu; Shu Yuan; Tingting Zhu; Xiaorong Wang; Youhua Ran; Sien Li; Chunyu Wang; Fei Jiang

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Abstract: N/A

18. A Gridded Inventory of Annual 2012-2018 U.S. Anthropogenic Methane Emissions

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Keywords: N/A

Funding: American Association for the Advancement of Science, National Aeronautics and Space Administration

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Open Access: N/A

Abstract: N/A

19. Global terrestrial nitrogen uptake and nitrogen use efficiency

Authors: Yunke Peng (Department of Environmental Systems Science ETH, Universitätsstrasse 2 Zurich Switzerland, Swiss Federal Institute for Forest Snow and Landscape Research WSL Birmensdorf Switzerland); Iain Colin Prentice (Imperial College London, Macquarie University, Tsinghua University); Keith J. Bloomfield (Imperial College London); Matteo Campioli (University of Antwerp); Zhiwen Guo (Beijing Forestry University); Yuanfeng Sun (Peking University); Di Tian (Department of Environmental Systems Science ETH, Universitätsstrasse 2 Zurich Switzerland, Beijing Forestry University, Swiss Federal Institute for Forest Snow and Landscape Research WSL Birmensdorf Switzerland); Xiangping Wang (Beijing Forestry University); Sara Vicca (University of Antwerp); Benjamin D. Stocker (Department of Environmental Systems Science ETH, Universitätsstrasse 2 Zurich Switzerland, University of Bern, Swiss Federal Institute for Forest Snow and Landscape Research WSL Birmensdorf Switzerland)

Journal: Journal of Ecology

Publisher: Wiley

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Abstract: Plant biomass production (BP), nitrogen uptake (Nup) and their ratio, and nitrogen use efficiency (NUE) must be quantified to understand how nitrogen (N) cycling constrains terrestrial carbon (C) uptake. But the controls of key plant processes determining Nup and NUE, including BP, C and N allocation, tissue C:N ratios and N resorption efficiency (NRE), remain poorly known. We compiled measurements from 804 forest and grassland sites and derived regression models for each of these processes with growth temperature, vapour pressure deficit, stand age, soil C:N ratio, fAPAR (remotely sensed fraction of photosynthetically active radiation absorbed by green vegetation) and growing-season average daily incident photosynthetic photon flux density (gPPFD; effectively the seasonal concentration of light availability, which increases polewards) as predictors. An empirical model for leaf N was based on optimal photosynthetic capacity (a function of gPPFD and climate) and observed leaf mass per area. The models were used to produce global maps of Nup and NUE. Global BP was estimated as 72 Pg C/year; Nup as 950 Tg N/year; and NUE as 76 g C/g N. Forest BP was found to increase with growth temperature and fAPAR and to decrease with stand age, soil C:N ratio and gPPFD. Forest NUE is controlled primarily by climate through its effect on C allocation--especially to leaves, being richer in N than other tissues. NUE is greater in colder climates, where N is less readily available, because below-ground allocation is increased. NUE is also greater in drier climates because

leaf allocation is reduced. NRE is enhanced (further promoting NUE) in both cold and dry climates. Synthesis. These findings can provide observationally based benchmarks for model representations of C-N cycle coupling. State-of-the-art vegetation models in the TRENDY ensemble showed variable performance against these benchmarks, and models including coupled C-N cycling produced relatively poor simulations of Nup and NUE.

20. A decrease in the age of respired carbon from the terrestrial biosphere and increase in the asymmetry of its distribution

Authors: Carlos A. Sierra (Max Planck Institute for Biogeochemistry); Gregory R. Quetin (Department of Earth System Science, Stanford University, Stanford, CA 94305, USA, University of California Santa Barbara); Holger Metzler (Swedish University of Agricultural Sciences, Max Planck Institute for Biogeochemistry); Markus Müller (Northern Arizona University, Max Planck Institute for Biogeochemistry)

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Open Access: N/A

Abstract: We provide here a model-based estimate of the transit time of carbon through the terrestrial biosphere, since the time of carbon uptake through photosynthesis until its release through respiration. We explored the consequences of increasing productivity versus increasing respiration rates on the transit time distribution and found that while higher respiration rates induced by higher temperature increase the transit time because older carbon is respired, increases in productivity cause a decline in transit times because more young carbon is available to supply increased metabolism. The combined effect of increases in temperature and productivity results in a decrease in transit times, with the productivity effect dominating over the respiration effect. By using an ensemble of simulation trajectories from the Carbon Data Model Framework (CARDAMOM), we obtained time-dependent transit time distributions incorporating the twentieth century global change. In these simulations, transit time declined over the twentieth century, suggesting an increased productivity effect that augmented the amount of respired young carbon, but also increasing the release of old carbon from high latitudes. The transit time distribution of carbon becomes more asymmetric over time, with more carbon transiting faster through tropical and temperate regions, and older carbon being respired from high latitude regions. This article is part of the Theo Murphy meeting issue 'Radiocarbon in the Anthropocene'.

21. Isolating the effects of land use and functional variation on Yucatán's forest biomass under global change

Authors: Stephanie P. George-Chacon; T. Luke Smallman; Juan Manuel Dupuy; José Luis Hernández-Stefanoni; David T. Milodowski; Mathew Williams

Journal: Frontiers in Forests and Global Change

Publisher: Frontiers Media SA

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Abstract: Tropical forests hold large stocks of carbon in biomass and face pressures from changing climate and anthropogenic disturbance. Forests' capacity to store biomass under future conditions and accumulate biomass during regrowth after clearance are major knowledge gaps. Here we use chronosequence data, satellite observations and a C-cycle model to diagnose woody C dynamics in two dry forest ecotypes (semi-deciduous and semi-evergreen) in Yucatán, Mexico. Woody biomass differences between mature semi-deciduous (90 MgC ha⁻¹) and semi-evergreen (175 MgC ha⁻¹) forest landscapes are mostly explained by differences in climate (c. 60%), particularly temperature, humidity and soil moisture effects on production. Functional variation in foliar phenology, woody allocation, and wood turnover rate explained c. 40% of biomass differences between ecotypes. Modeling experiments explored varied forest clearance and regrowth cycles, under a range of climate and CO₂ change scenarios to 2100. Production and steady state biomass in both ecotypes were reduced by forecast warming and drying (mean biomass 2021-2100 reduced 16-19% compared to 2001-2020), but compensated by fertilisation from rising CO₂. Functional analysis indicates that trait adjustments amplify biomass losses by 70%. Experiments with disturbance and recovery across historically reported levels indicate reductions to mean forest biomass stocks over 2021-2100 similar in magnitude to climate impacts (10-19% reductions for disturbance with recovery). Forest disturbance without regrowth amplifies biomass loss by three- or four-fold. We conclude that vegetation functional differences across the Yucatán climate gradient have developed to limit climate risks. Climate change will therefore lead to functional adjustments for all forest types. These adjustments are likely to magnify biomass reductions caused directly by climate change over the coming century. However, the range of impacts of land use and land use change are as, or more, substantive than the totality of direct and indirect climate impacts. Thus the carbon storage of Yucatan's forests is highly vulnerable both to climate and land use and land use change. Our results here should be used to test and enhance land surface models use for dry forest carbon cycle assessment regionally and globally. A single plant functional type approach for modeling Yucatán's forests is not justified.

22. Inner Mongolia grasslands act as a weak regional carbon sink: A new estimation based on upscaling eddy covariance observations

Authors: Cuihai You; Yanbing Wang; Xingru Tan; Erqian Cui; Bingwei Zhang; Chenyu Bian; Boyu Chen; Mengzhen Xu; Xingguo Han; Jianyang Xia; Shiping Chen

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Abstract: N/A

23. Resolving the Carbon-Climate Feedback Potential of Wetland CO₂ and CH₄ Fluxes in Alaska

Authors: Shuang Ma (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jennifer D. Watts (Woodwell Climate Research Center Falmouth MA USA); Gregory R. Quetin (University of California Santa Barbara); Zona Donatella (Department of Biology San Diego State University San Diego CA USA); Eugénie S. Euskirchen (University of Alaska Fairbanks); Alexander J. Norton (Australian National University); Yi Yin (California Institute of Technology); Paul A. Levine (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Renato K. Braghiere (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John R. Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); David S. Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Keywords: CO₂, climatic sensitivities, global warming potential, northern high latitude, carbon-climate feedback (and 1 more)

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Open Access: N/A

Abstract: Boreal-Arctic regions are key stores of organic carbon (C) and play a major role in the greenhouse gas balance of high-latitude ecosystems. The carbon-climate (C-climate) feedback potential of northern high-latitude ecosystems remains poorly understood due to uncertainty in temperature and precipitation controls on carbon dioxide (CO₂) uptake and the decomposition of soil C into CO₂ and methane (CH₄) fluxes. While CH₄ fluxes account for a smaller component of the C balance, the climatic impact of CH₄ outweighs CO₂ (28-34 times larger global warming potential on a 100-year scale), highlighting the need to jointly resolve the climatic sensitivities of both CO₂ and CH₄. Here, we jointly constrain a terrestrial biosphere model with in situ CO₂ and CH₄ flux observations at seven eddy covariance sites using a data-model integration approach to resolve the integrated

environmental controls on land-atmosphere CO₂ and CH₄ exchanges in Alaska. Based on the combined CO₂ and CH₄ flux responses to climate variables, we find that 1970-present climate trends will induce positive C-climate feedback at all tundra sites, and negative C-climate feedback at the boreal and shrub fen sites. The positive C-climate feedback at the tundra sites is predominantly driven by increased CH₄ emissions while the negative C-climate feedback at the boreal site is predominantly driven by increased CO₂ uptake (80% from decreased heterotrophic respiration, and 20% from increased photosynthesis). Our study demonstrates the need for joint observational constraints on CO₂ and CH₄ biogeochemical processes--and their associated climatic sensitivities--for resolving the sign and magnitude of high-latitude ecosystem C-climate feedback in the coming decades.

24. Atmospheric CO₂ inversion reveals the Amazon as a minor carbon source caused by fire emissions, with forest uptake offsetting about half of these emissions

Authors: Luana S. Basso; Chris Wilson; Martyn P. Chipperfield; Graciela Tejada; Henrique L. G. Cassol; Egídio Arai; Mathew Williams; T. Luke Smallman; Wouter Peters; Stijn Naus; John B. Miller; Manuel Gloor

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Open Access: N/A

Abstract: . Tropical forests such as the Amazonian rainforests play an important role for climate, are large carbon stores and are a treasure of biodiversity. Amazonian forests have been exposed to large-scale deforestation and degradation for many decades. Deforestation declined between 2005 and 2012 but more recently has again increased with similar rates as in 2007-2008. The resulting forest fragments are exposed to substantially elevated temperatures in an already warming world. These temperature and land cover changes are expected to affect the forests, and an important diagnostic of their health and sensitivity to climate variation is their carbon balance. In a recent study based on CO₂ atmospheric vertical profile observations between 2010 and 2018, and an air column budgeting technique used to estimate fluxes, we reported the Amazon region as a carbon source to the atmosphere, mainly due to fire emissions. Instead of an air column budgeting technique, we use an inverse of the global atmospheric transport model, TOMCAT, to assimilate CO₂ observations from Amazon vertical profiles and global flask measurements. We thus estimate inter- and intra-annual variability in the carbon fluxes, trend over time and controls for the period of 2010-2018. This is the longest period covered by a Bayesian inversion of these atmospheric CO₂ profile observations to date. Our analyses indicate that the Amazon is a small net source of carbon to the atmosphere (mean 2010-2018 = 0.13 ± 0.17 Pg C yr⁻¹, where 0.17 is the 1σ uncertainty), with the majority of the emissions coming from the eastern region (77 % of total Amazon emissions). Fire is the primary driver of the Amazonian source (0.26 ± 0.13 Pg C yr⁻¹), while forest carbon uptake removes around half of the fire emissions to the atmosphere (-0.13 ± 0.20 Pg C yr⁻¹). The largest net carbon sink was observed in the western-central Amazon region (72 % of the fire emissions). We find larger carbon emissions during the extreme drought years (such as 2010, 2015 and 2016), correlated with increases in

temperature, cumulative water deficit and burned area. Despite the increase in total carbon emissions during drought years, we do not observe a significant trend over time in our carbon total, fire and net biome exchange estimates between 2010 and 2018. Our analysis thus cannot provide clear evidence for a weakening of the carbon uptake by Amazonian tropical forests.

25. Scale variance in the carbon dynamics of fragmented, mixed-use landscapes estimated using model-data fusion

Authors: David T. Milodowski; T. Luke Smallman; Mathew Williams

Journal: Biogeosciences

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Abstract: . Many terrestrial landscapes are heterogeneous. Mixed land cover and land use generate a complex mosaic of fragmented ecosystems at fine spatial resolutions with contrasting ecosystem stocks, traits, and processes, each differently sensitive to environmental and human factors. Representing spatial complexity within terrestrial ecosystem models is a key challenge for understanding regional carbon dynamics, their sensitivity to environmental gradients, and their resilience in the face of climate change. Heterogeneity underpins this challenge due to the trade-off between the fidelity of ecosystem representation within modelling frameworks and the computational capacity required for fine-scale model calibration and simulation. We directly address this challenge by quantifying the sensitivity of simulated carbon fluxes in a mixed-use landscape in the UK to the spatial resolution of the model analysis. We test two different approaches for combining Earth observation (EO) data into the CARDAMOM model-data fusion (MDF) framework, assimilating time series of satellite-based EO-derived estimates of ecosystem leaf area and biomass stocks to constrain estimates of model parameters and their uncertainty for an intermediate complexity model of the terrestrial C cycle. In the first approach, ecosystems are calibrated and simulated at pixel level, representing a "community average" of the encompassed land cover and management. This represents our baseline approach. In the second, we stratify each pixel based on land cover (e.g. coniferous forest, arable/pasture) and calibrate the model independently using EO data specific to each stratum. We test the scale dependence of these approaches for grid resolutions spanning 1 to 0.05 km over a mixed-land-use region of the UK. Our analyses indicate that spatial resolution matters for MDF. Under the community average baseline approach biological C fluxes (gross primary productivity, Reco) simulated by CARDAMOM are relatively insensitive to resolution. However, disturbance fluxes exhibit scale variance that increases with greater landscape fragmentation and for coarser model domains. In contrast, stratification of assimilated data based on fine-resolution land use distributions resolved the resolution dependence, leading to disturbance fluxes that were 40 %-100 % higher than the baseline experiments. The differences in the simulated disturbance fluxes result in estimates of the terrestrial carbon balance in the stratified experiment that suggest a weaker C sink compared to the baseline

experiment. We also find that stratifying the model domain based on land use leads to differences in the retrieved parameters that reflect variations in ecosystem function between neighbouring areas of contrasting land use. The emergent differences in model parameters between land use strata give rise to divergent responses to future climate change. Accounting for fine-scale structure in heterogeneous landscapes (e.g. stratification) is therefore vital for ensuring the ecological fidelity of large-scale MDF frameworks. The need for stratification arises because land use places strong controls on the spatial distribution of carbon stocks and plant functional traits and on the ecological processes controlling the fluxes of C through landscapes, particularly those related to management and disturbance. Given the importance of disturbance to global terrestrial C fluxes, together with the widespread increase in fragmentation of forest landscapes, these results carry broader significance for the application of MDF frameworks to constrain the terrestrial C balance at regional and national scales.

26. Forest productivity recovery or collapse? Model-data integration insights on drought-induced tipping points

Authors: J. Au (University of California Davis); A. A. Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); N. C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); R. M. Deans (University of Edinburgh); C. Y. S. Wong (University of California Davis); B. Z. Houlton (Department of Ecology and Evolutionary Biology and Department of Global Development Cornell University Ithaca New York USA); T. S. Magney (University of California Davis)

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Abstract: More frequent and severe droughts are driving increased forest mortality around the globe. We urgently need to describe and predict how drought affects forest carbon cycling and identify thresholds of environmental stress that trigger ecosystem collapse. Quantifying the effects of drought at an ecosystem level is complex because dynamic climate-plant relationships can cause rapid and/or prolonged shifts in carbon balance. We employ the CARbon DATA MOdel fraMework (CARDAMOM) to investigate legacy effects of drought on forest carbon pools and fluxes. Our Bayesian model-data fusion approach uses tower observed meteorological forcing and carbon fluxes to determine the response and sensitivity of aboveground and belowground ecological processes associated with the 2012-2015 California drought. Our study area is a mid-montane mixed conifer forest in the Southern Sierras. CARDAMOM constrained with gross primary productivity (GPP) estimates covering 2011-2017 show a ~75% reduction in GPP, compared to negligible GPP change when constrained with 2011 only. Precipitation across 2012-2015 was 45% (474 mm) lower than the historical average and drove a cascading depletion in soil moisture and carbon pools (foliar, labile, roots, and litter). Adding 157 mm during an especially stressful year (2014, annual rainfall = 293 mm) led to a smaller depletion of water and carbon pools, steering the ecosystem away from a state of GPP tipping-point collapse to recovery. We present novel process-driven insights that demonstrate the sensitivity of GPP collapse to ecosystem foliar carbon and soil moisture states--showing that the full extent of GPP response takes several years to arise. Thus, long-term changes in soil moisture and carbon pools can provide a mechanistic link between drought and forest mortality. Our study provides an example for how key precipitation threshold ranges can influence forest productivity, making them useful for monitoring and predicting forest mortality events.

27. Biogeographic pattern of living vegetation carbon turnover time in mature forests across continents

Authors: Kailiang Yu (Le Laboratoire des Sciences du Climat et de l'Environnement IPSL-LSCECEA/CNRS/UVSQ Saclay Gif-sur-Yvette France, University of Montana); Philippe Ciais (Le Laboratoire des Sciences du Climat et de l'Environnement IPSL-LSCECEA/CNRS/UVSQ Saclay Gif-sur-Yvette France, The Cyprus Institute); Anthony A. Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Jinsong Wang (Chinese Academy of Sciences); Zhihua Liu (University of Montana); Han Y. H. Chen (Lakehead University); Yilong Wang (Chinese Academy of Sciences); Yizhao Chen (Nanjing Forestry University); Ashley P. Ballantyne (Le Laboratoire des Sciences du Climat et de l'Environnement IPSL-LSCECEA/CNRS/UVSQ Saclay Gif-sur-Yvette France, University of Montana)

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Open Access: N/A

Abstract: AimTheoretically, woody biomass turnover time (τ) quantified using outflux (i.e. tree mortality) predicts biomass dynamics better than using influx (i.e. productivity). This study aims at using forest inventory data to empirically test the outflux approach and generate a spatially explicit understanding of woody in mature forests. We further compared woody estimates with dynamic global vegetation models (DGVMs) and with a data assimilation product of C stocks and fluxes--CARDAMOM. LocationContinents. Time PeriodHistoric from 1951 to 2018. Major Taxa StudiedTrees and forests. MethodsWe compared the approaches of using outflux versus influx for estimating woody and predicting biomass accumulation rates. We investigated abiotic and biotic drivers of spatial woody and generated a spatially explicit map of woody at a 0.25-degree resolution across continents using machine learning. We further examined whether six DGVMs and CARDAMOM generally captured the observational pattern of woody. ResultsWoody quantified by the outflux approach better (with R^2 0.4-0.5) predicted the biomass accumulation rates than the influx approach (with R^2 0.1-0.4) across continents. We found large spatial variations of woody for mature forests, with highest values in temperate forests (98.8 +/- 2.6 y) followed by boreal forests (73.9 +/- 3.6 y) and tropical forests. The map of woody extrapolated from plot data showed higher values in wetter eastern and pacific coast USA, Africa and eastern Amazon. Climate (temperature and aridity index) and vegetation structure (tree density and forest age) were the dominant drivers of woody across continents. The highest woody in temperate forests was not captured by either DGVMs or

CARDAMOM. Main Conclusions Our study empirically demonstrated the preference of using outflux over influx to estimate woody for predicting biomass accumulation rates. The spatially explicit map of woody and the underlying drivers provide valuable information to improve the representation of forest demography and carbon turnover processes in DGVMs.

28. Improved Constraints on the Recent Terrestrial Carbon Sink Over China by Assimilating OCO-2 XCO₂ Retrievals

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Keywords: atmospheric inversion, China, terrestrial carbon sink, Orbit Carbon Observatory 2 (OCO-2), atmospheric CO₂ column concentration

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Abstract: The magnitude and distribution of China's terrestrial carbon sink remain uncertain due to insufficient observational constraints; satellite column-average dry-air mole fraction carbon dioxide (XCO₂) retrievals may fill some of this gap. Here, we estimate China's carbon sink using atmospheric inversions of the Orbiting Carbon Observatory 2 (OCO-2) XCO₂ retrievals within different platforms, including the Global Carbon Assimilation System (GCAS) v2, the Copernicus Atmosphere Monitoring Service, and the OCO-2 Model Inter-comparison Project (MIP). We find that they consistently place the largest net biome production (NBP) in the south on an annual basis compared to the northeast and other main agricultural areas during peak growing season, coinciding well with the distribution of forests and crops, respectively. Moreover, the mean seasonal cycle amplitude of NBP in OCO-2 inversions is obviously larger than that of biosphere model simulations and slightly greater than surface CO₂ inversions. More importantly, the mean seasonal cycle of the OCO-2 inversions is well constrained in the temperate, tropical, and subtropical monsoon climate zones, with better inter-model consistency at a sub-regional scale compared to in situ inversions and biosphere model simulations. In addition, the

OCO-2 inversions estimate the mean annual NBP in China for 2015-2019 to be between 0.34 (GCASv2) and 0.47 ± 0.16 PgC/yr (median $\pm$ std; OCO-2 v10 MIP), and indicate the impacts of climate extremes (e.g., the 2019 drought) on the interannual variations of NBP. Our results suggest that assimilating OCO-2 XCO₂ retrievals is crucial for improving our understanding of China's terrestrial carbon sink regime.

29. Improved process representation of leaf phenology significantly shifts climate sensitivity of ecosystem carbon balance

Authors: Alexander J. Norton; A. Anthony Bloom; Nicholas C. Parazoo; Paul A. Levine; Shuang Ma; Renato K. Braghiere; T. Luke Smallman

Journal: Biogeosciences

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Abstract: . Terrestrial carbon cycle models are routinely used to determine the response of the land carbon sink under expected future climate change, yet these predictions remain highly uncertain. Increasing the realism of processes in these models may help with predictive skill, but any such addition should be confronted with observations and evaluated in the context of the aggregate behavior of the carbon cycle. Here, two formulations for leaf area index (LAI) phenology are coupled to the same terrestrial biosphere model: one is climate agnostic, and the other incorporates direct environmental controls on both timing and growth. Each model is calibrated simultaneously to observations of LAI, net ecosystem exchange (NEE), and biomass using the CARbon DATA-MODEL fraMework (CARDAMOM) and validated against withheld data, including eddy covariance estimates of gross primary productivity (GPP) and ecosystem respiration (Re) across six ecosystems from the tropics to high latitudes. Both model formulations show similar predictive skill for LAI and NEE. However, with the addition of direct environmental controls on LAI, the integrated model explains 22 % more variability in GPP and Re and reduces biases in these fluxes by 58 % and 77 %, respectively, while also predicting more realistic annual litterfall rates due to changes in carbon allocation and turnover. We extend this analysis to evaluate the inferred climate sensitivity of LAI and NEE with the new model and show that the added complexity shifts the sign, magnitude, and seasonality of NEE sensitivity to precipitation and temperature. This highlights the benefit of process complexity when inferring underlying processes from Earth observations and representing the climate response of the terrestrial carbon cycle.

30. National CO2 budgets (2015-2020) inferred from atmospheric CO2 observations in support of the global stocktake

Authors: Brendan Byrne; David F. Baker; Sourish Basu; Michael Bertolacci; Kevin W. Bowman; Dustin Carroll; Abhishek Chatterjee; Frédéric Chevallier; Philippe Ciais; Noel Cressie; David Crisp; Sean Crowell; Feng Deng; Zhu Deng; Nicholas M. Deutsch; Manvendra K. Dubey; Sha Feng; Omaira E. García; David W. T. Griffith; Benedikt Herkommer; Lei Hu; Andrew R. Jacobson; Rajesh Janardanan; Sujong Jeong; Matthew S. Johnson; Dylan B. A. Jones; Rigel Kivi; Junjie Liu; Zhiqiang Liu; Shamil Maksyutov; John B. Miller; Scot M. Miller; Isamu Morino; Justus Notholt; Tomohiro Oda; Christopher W. O'Dell; Young-Suk Oh; Hirofumi Ohyama; Prabir K. Patra; Hélène Peiro; Christof Petri; Sajeev Philip; David F. Pollard; Benjamin Poulter; Marine Remaud; Andrew Schuh; Mahesh K. Sha; Kei Shiomi; Kimberly Strong; Colm Sweeney; Yao Té; Hanqin Tian; Voltaire A. Velazco; Mihalis Vrekoussis; Thorsten Warneke; John R. Worden; Debra Wunch; Yuanzhi Yao; Jeongmin Yun; Andrew Zammit-Mangion; Ning Zeng

Journal: Earth System Science Data

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Abstract: . Accurate accounting of emissions and removals of CO₂ is critical for the planning and verification of emission reduction targets in support of the Paris Agreement. Here, we present a pilot dataset of country-specific net carbon exchange (NCE; fossil plus terrestrial ecosystem fluxes) and terrestrial carbon stock changes aimed at informing countries' carbon budgets. These estimates are based on "top-down" NCE outputs from the v10 Orbiting Carbon Observatory (OCO-2) modeling intercomparison project (MIP), wherein an ensemble of inverse modeling groups conducted standardized experiments assimilating OCO-2 column-averaged dry-air mole fraction (XCO₂) retrievals (ACOS v10), in situ CO₂ measurements or combinations of these data. The v10 OCO-2 MIP NCE estimates are combined with "bottom-up" estimates of fossil fuel emissions and lateral carbon fluxes to estimate changes in terrestrial carbon stocks, which are impacted by anthropogenic and natural drivers. These flux and stock change estimates are reported annually (2015-2020) as both a global 1° x 1° gridded dataset and a country-level dataset and are available for download from the Committee on Earth Observation Satellites' (CEOS) website:<https://doi.org/10.48588/npf6-sw92> (Byrne et al., 2022). Across the v10 OCO-2 MIP experiments, we obtain increases in the ensemble median terrestrial carbon stocks of 3.29-4.58 Pg CO₂ yr⁻¹ (0.90-1.25 Pg C yr⁻¹). This is a result of broad increases in

terrestrial carbon stocks across the northern extratropics, while the tropics generally have stock losses but with considerable regional variability and differences between v10 OCO-2 MIP experiments. We discuss the state of the science for tracking emissions and removals using top-down methods, including current limitations and future developments towards top-down monitoring and verification systems.

31. Optimizing the Isoprene Emission Model MEGAN With Satellite and Ground-Based Observational Constraints

Authors: Christian A. DiMaria (University of Toronto); Dylan B. A. Jones (University of Toronto); Helen Worden (National Center for Atmospheric Research); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Trissevgeni Stavrakou (Royal Belgian Institute for Space Aeronomy); Kazuyuki Miyazaki (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); John Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Alex Guenther (University of California Irvine); Chinmoy Sarkar (University of California Irvine); Roger Seco (Institute of Environmental Assessment and Water Research (IDAEA-CSIC) Barcelona Spain); Jeong-Hoo Park (Air Quality Research Division National Institute of Environmental Research Incheon Republic of Korea); Julio Tota (Julio Tota Instituto de engenharia e Geociências Universidade Federal do Oeste do Pará UFOPA Santarém Brazil); Eliane Gomes Alves (Max Planck Institute for Biogeochemistry); Valerio Ferracci (Cranfield University)

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Abstract: Isoprene is a hydrocarbon emitted in large quantities by terrestrial vegetation. It is a precursor to several air quality and climate pollutants including ozone. Emission rates vary with plant species and environmental conditions. This variability can be modeled using the Model of Emissions of Gases and Aerosols from Nature (MEGAN). MEGAN parameterizes isoprene emission rates as a vegetation-specific standard rate which is modulated by scaling factors that depend on meteorological and environmental driving variables. Recent experiments have identified large uncertainties in the MEGAN temperature response parameterization, while the emission rates under standard conditions are poorly constrained in some regions due to a lack of representative measurements and uncertainties in landcover. In this study, we use Bayesian model-data fusion to optimize the MEGAN temperature response and standard emission rates using satellite- and ground-based observational constraints. Optimization of the standard emission rate with satellite constraints reduced model biases but was highly sensitive to model input errors and drought stress and was found to be inconsistent with ground-based constraints at an Amazonian field site, reflecting large uncertainties in the satellite-based

emissions. Optimization of the temperature response with ground-based constraints increased the temperature sensitivity of the model by a factor of five at an Amazonian field site but had no impact at a UK field site, demonstrating significant ecosystem-dependent variability of the isoprene emission temperature sensitivity. Ground-based measurements of isoprene across a wide range of ecosystems will be key for obtaining an accurate representation of isoprene emission temperature sensitivity in global biogeochemical models.

32. Attributing Past Carbon Fluxes to CO₂ and Climate Change: Respiration Response to CO₂ Fertilization Shifts Regional Distribution of the Carbon Sink

Authors: Gregory R. Quetin (University of California Santa Barbara, Department of Earth System Science Stanford University Stanford CA USA); Caroline A. Famiglietti (Department of Earth System Science Stanford University Stanford CA USA); Nathan C. Dadap (Department of Earth System Science Stanford University Stanford CA USA); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Noah S. Diffenbaugh (Department of Earth System Science Stanford University Stanford CA USA, Woods Institute for the Environment Stanford University Stanford CA USA); Junjie Liu (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Anna T. Trugman (University of California Santa Barbara); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA, Woods Institute for the Environment Stanford University Stanford CA USA)

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Abstract: Over the past century, increased atmospheric CO₂ concentrations have enhanced photosynthesis through CO₂ fertilization across the globe. However, the increased growth has also led to greater respiration rates--both from vegetation (autotrophic respiration) and through the breakdown of plant litter and soil organic matter (heterotrophic respiration). The resulting change in carbon flux--and its spatial distribution--that can be attributed to increasing CO₂ and climate change remains unknown. We used the Carbon Data Model Framework, a model-data fusion system that assimilates global observations from satellites and other sources to create an ensemble of observationally constrained carbon cycle representations, to determine the photosynthesis and respiration fluxes that can be attributed to increased atmospheric CO₂ and associated climate change from 1920 to 2015. Across the globe, the response of photosynthesis and respiration to atmospheric CO₂ dominates their response to climate alone. The regional distribution of the carbon sink attributable to climate change and CO₂ is strongly influenced by the 'loss ratio of carbon gained'--the fraction of enhanced photosynthesis that is lost to respiration. While the wet tropics' attributable photosynthesis flux is 1.4 times larger than that of the temperate region, the attributable flux of net carbon uptake is actually 1.25

larger in the temperate region, due to the wet tropics' greater heterotrophic respiration response to enhanced plant growth. At the global scale, the loss ratio of carbon gained is $83 \pm 0.6\%$. Our results highlight the importance of the respiration responses to enhanced plant growth in regulating the land carbon sink.

33. Global net biome CO₂ exchange predicted comparably well using parameter-environment relationships and plant functional types

Authors: Caroline A. Famiglietti (Department of Earth System Science Stanford University Stanford California USA); Matthew Worden (Department of Earth System Science Stanford University Stanford California USA); Gregory R. Quetin (Department of Geography University of California at Santa Barbara Santa Barbara California USA); T. Luke Smallman (University of Edinburgh); Uma Dayal (Department of Earth System Science Stanford University Stanford California USA); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mathew Williams (University of Edinburgh); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford California USA)

Journal: Global Change Biology

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Keywords: N/A

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Abstract: Accurate estimation and forecasts of net biome CO₂ exchange (NBE) are vital for understanding the role of terrestrial ecosystems in a changing climate. Prior efforts to improve NBE predictions have predominantly focused on increasing models' structural realism (and thus complexity), but parametric error and uncertainty are also key determinants of model skill. Here, we investigate how different parameterization assumptions propagate into NBE prediction errors across the globe, pitting the traditional plant functional type (PFT)-based approach against a novel top-down, machine learning-based "environmental filtering" (EF) approach. To do so, we simulate these contrasting methods for parameter assignment within a flexible model-data fusion framework of the terrestrial carbon cycle (CARDAMOM) at a global scale. In the PFT-based approach, model parameters from a small number of select locations are applied uniformly within regions sharing similar land cover characteristics. In the EF-based approach, a pixel's parameters are predicted based on underlying relationships with climate, soil, and canopy properties. To isolate the role of parametric from structural uncertainty in our analysis, we benchmark the resulting PFT-based and EF-based NBE predictions with estimates from CARDAMOM's Bayesian optimization approach (whereby "true" parameters consistent with a suite of data constraints are retrieved on a pixel-by-pixel basis). When considering the mean absolute error of NBE predictions across time, we find that the EF-based approach matches or outperforms the PFT-based approach at 55% of pixels--a narrow majority. However, NBE estimates from the EF-based approach are susceptible to compensation between errors in component flux

predictions and predicted parameters can align poorly with the assumed "true" values. Overall, though, the EF-based approach is comparable to conventional approaches and merits further investigation to better understand and resolve these limitations. This work provides insight into the relationship between terrestrial biosphere model performance and parametric uncertainty, informing efforts to improve model parameterization via PFT-free and trait-based approaches.

34. Evaluation of soil carbon simulation in CMIP6 Earth system models

Authors: Rebecca M. Varney; Sarah E. Chadburn; Eleanor J. Burke; Peter M. Cox

Journal: Biogeosciences

Publisher: Copernicus GmbH

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Open Access: N/A

Abstract: . The response of soil carbon represents one of the key uncertainties in future climate change. The ability of Earth system models (ESMs) to simulate present-day soil carbon is therefore vital for reliably estimating global carbon budgets required for Paris Agreement targets. In this study CMIP6 ESMs are evaluated against empirical datasets to assess the ability of each model to simulate soil carbon and related controls: net primary productivity (NPP) and soil carbon turnover time (taus). Comparing CMIP6 with the previous generation of models (CMIP5), a lack of consistency in modelled soil carbon remains, particularly the underestimation of northern high-latitude soil carbon stocks. There is a robust improvement in the simulation of NPP in CMIP6 compared with CMIP5; however, an unrealistically high correlation with soil carbon stocks remains, suggesting the potential for an overestimation of the long-term terrestrial carbon sink. Additionally, the same improvements are not seen in the simulation of taus. These results suggest that much of the uncertainty associated with modelled soil carbon stocks can be attributed to the simulation of below-ground processes, and greater emphasis is required on improving the representation of below-ground soil processes in future developments of models. These improvements would help to reduce the uncertainty in projected carbon release from global soils under climate change and to increase confidence in the carbon budgets associated with different levels of global warming.

35. Evaluating carbon and water fluxes and stocks in Brazil under changing climate and refined regional scenarios for changes in land use

Authors: Aline Anderson de Castro; Celso von Randow; Rita de Cássia Silva von Randow; Francisco Gilney Silva Bezerra

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Publisher: Frontiers Media SA

DOI: 10.3389/fclim.2022.941900

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Abstract: Climate change and land-use change can alter the role of natural vegetation as a sink or source of atmospheric carbon. In this work, we evaluate the response of water and carbon fluxes and stocks in Brazilian biomes as a proxy for ecosystem services of regional climate regulation under two contrasting future scenarios: a sustainable development scenario, where some deforested areas are restored by vegetation regrowth combined with a low representative concentration pathway, and a pessimistic scenario, where there is still high deforestation rates and strong climate change. We used refined regional scenarios for land-use change in Brazil, together with climate projections of the HADGEM2-ES model for RCPs 2.6 and 8.5 to drive a land surface model and assess possible future impacts in surface fluxes. Our results show that drying climate and shifts of natural vegetation into anthropogenic land use might shift part of upperstory biomass into understory biomass, which can be more vulnerable to dry events. The simulations also show that climate change appears to drive most of the water balance changes compared to land-use change, especially over the Amazon.

36. The carbon budget of the managed grasslands of Great Britain - informed by earth observations

Authors: Vasileios Myrgiotis; Thomas Luke Smallman; Mathew Williams

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-19-4147-2022

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Abstract: . Grasslands cover around two-thirds of the agricultural land area of Great Britain (GB) and are important reservoirs of organic carbon (C). Direct assessments of the C balance of grasslands require continuous monitoring of C pools and fluxes, which is only possible at a small number of experimental sites. By relying on our quantitative understanding of ecosystem C biogeochemistry we develop models of grassland C dynamics and use them to estimate grassland C balance at various scales. Model-based estimation of the C budget of individual fields and across large domains is made complex by the spatial and temporal variability in climate and soil conditions, as well as in livestock grazing, grass cutting and other management activities. In this context, earth observations (EOs) provide subfield-resolution proxy data on the state of grassland canopies, allowing us to infer information about vegetation management, to apply observational constraints to the simulated ecosystems and, thus, to mitigate the effects of model input data uncertainty. Here, we show the potential of model-data fusion (MDF) methods to provide robust analyses of C dynamics in managed grasslands across GB. We combine EO data and biogeochemical modelling by implementing a probabilistic MDF algorithm to (1) assimilate leaf area index (LAI) times series (Sentinel-2); (2) infer defoliation instances (grazing, cutting); and (3) simulate livestock grazing, grass cutting, and C allocation and C exchanges with the atmosphere. The algorithm uses the inferred information on grazing and cutting to drive the model's C removals-and-returns module, according to which $\approx 1/3$ of C in grazed biomass returns to the soil as manure (other inputs of manure not considered) and C in cut grass is removed from the system (downstream C emissions not considered). Spatial information on soil C stocks is obtained from the SoilGrids dataset. The MDF algorithm was applied for 2017-2018 to generate probabilistic estimates of C pools and fluxes at 1855 fields sampled from across GB. The algorithm was able to effectively assimilate the Sentinel-2-based LAI time series (overlap = 80 %, RMSE = 1.1 m² m⁻², bias = 0.35 m² m⁻²) and predict livestock densities per area that correspond with independent agricultural census-based data ($r = 0.68$, RMSE = 0.45 LU ha⁻¹, bias = -0.06 LU ha⁻¹). The mean total removed biomass across all simulated fields was 6 (+/-1.8) t DM ha⁻¹ yr⁻¹. The simulated grassland ecosystems were on average C sinks in 2017 and 2018; the net biome exchange

(NBE) was -191 ± 81 (2017) and -49 ± 69 gC m⁻² yr⁻¹ (2018). Our results show that the 2018 European summer drought reduced the strength of C sinks in GB grasslands and led to a 9-fold increase in the number of fields that were annual C sources (NBE > 0) in 2018 (18 % of fields) compared to 2017 (2 % of fields). The field-scale analysis showed that management in the form of timing, intensity and type of defoliation were key determinants of the C balance of managed grasslands, with cut fields acting as weaker C sinks compared to grazed fields. Nevertheless, extreme weather, such as prolonged droughts, can convert grassland C sinks to sources.

37. Quantifying Northern High Latitude Gross Primary Productivity (GPP) Using Carbonyl Sulfide (OCS)

Authors: Le Kuai (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Mingjie Shi (Pacific Northwest National Laboratory); Charles E. Miller (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Ian Baker (Colorado State University); Anthony A. Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Zhao-Cheng Zeng (University of California Los Angeles); Roisin Commane (Columbia University); Stephen A. Montzka (NOAA Global Monitoring Laboratory Boulder CO USA); Joe Berry (Stanford University Stanford CA USA); Colm Sweeney (NOAA Global Monitoring Laboratory Boulder CO USA); John B. Miller (NOAA Global Monitoring Laboratory Boulder CO USA); Yuk L. Yung (California Institute of Technology, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology)

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Abstract: The northern high latitude (NHL, 40 degreesN to 90 degreesN) is where the second peak region of gross primary productivity (GPP) other than the tropics. The summer NHL GPP is about 80% of the tropical peak, but both regions are still highly uncertain (Norton et al. 2019, <https://doi.org/10.5194/bg-16-3069-2019>). Carbonyl sulfide (OCS) provides an important proxy for photosynthetic carbon uptake. Here we optimize the OCS plant uptake fluxes across the NHL by fitting atmospheric concentration simulation with the GEOS-CHEM global transport model to the aircraft profiles acquired over Alaska during NASA's Carbon in Arctic Reservoirs Vulnerability Experiment (2012-2015). We use the empirical biome-specific linear relationship between OCS plant uptake flux and GPP to derive the six plant uptake OCS fluxes from different GPP data. Such GPP-based fluxes are used to drive the concentration simulations. We evaluate the simulations against the independent observations at two ground sites of Alaska. The optimized OCS fluxes suggest the NHL plant uptake OCS flux of -247 Gg S year⁻¹, about 25% stronger than the ensemble mean of the six GPP-based OCS

fluxes. GPP-based OCS fluxes systematically underestimate the peak growing season across the NHL, while a subset of models predict early start of season in Alaska, consistent with previous studies of net ecosystem exchange. The OCS optimized GPP of 34 PgC yr⁻¹ for NHL is also about 25% more than the ensembles mean from six GPP data. Further work is needed to fully understand the environmental and biotic drivers and quantify their rate of photosynthetic carbon uptake in Arctic ecosystems.

38. Global Carbon Cycle Data Assimilation Using Earth Observation

Authors: Mathew Williams

Journal: Land Carbon Cycle Modeling

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Abstract: N/A

39. Challenges Regionalizing Methane Emissions Using Aquatic Environments in the Amazon Basin as Examples

Authors: John M. Melack; Luana S. Basso; Ayan S. Fleischmann; Santiago Botía; Mingyang Guo; Wencai Zhou; Pedro M. Barbosa; Joao H.F. Amaral; Sally MacIntyre

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License: CC BY 4.0

Open Access: N/A

Abstract: Key challenges to regionalization of methane fluxes in the Amazon basin are the large seasonal variation in inundated areas and habitats, the wide variety of aquatic ecosystems throughout the Amazon basin, and the variability in methane fluxes in time and space. Based on available measurements of methane emission and areal extent, seven types of aquatic systems are considered: streams and rivers, lakes, seasonally flooded forests, seasonally flooded savannas and other interfluvial wetlands, herbaceous plants on riverine floodplains, peatlands, and hydroelectric reservoirs. We evaluate the adequacy of sampling and of field methods plus atmospheric measurements, as applied to the Amazon basin, summarize published fluxes and regional estimates using bottom-up and top-down approaches, and discuss current understanding of biogeochemical and physical processes in Amazon aquatic environments and their incorporation into mechanistic and statistical models. Recommendations for further study in the Amazon basin and elsewhere include application of new remote sensing techniques, increased sampling frequency and duration, experimental studies to improve understanding of biogeochemical and physical processes, and development of models appropriate for hydrological and ecological conditions.

40. Challenges in Scaling Up Greenhouse Gas Fluxes: Experience From the UK Greenhouse Gas Emissions and Feedbacks Program

Authors: Peter Levy (Centre of Ecology and Hydrology Penicuik UK); Robert Clement (University of Edinburgh); Nick Cowan (Centre of Ecology and Hydrology Penicuik UK); Ben Keane (University of York); Vasilis Myrgiotis (University of Edinburgh); Marcel van Oijen (Centre of Ecology and Hydrology Penicuik UK); T. Luke Smallman (University of Edinburgh); Sylvia Toet (University of York); Mathew Williams (University of Edinburgh)

Journal: Journal of Geophysical Research: Biogeosciences

Publisher: American Geophysical Union (AGU)

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Volume/Issue: Volume 127, Issue 5

References: 85

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

License: CC BY 4.0

Open Access: N/A

Abstract: The role of greenhouse gases (GHGs) in global climate change is now well recognized and there is a clear need to measure emissions and verify the efficacy of mitigation measures. To this end, reliable estimates are needed of the GHG balance at the national scale and over long time periods, but these estimates are difficult to make accurately. Because measurement techniques are generally restricted to relatively small spatial and temporal scales, there is a fundamental problem in translating these into long-term estimates on a regional scale. The key challenge lies in spatial and temporal upscaling of short-term, point observations to estimate large-scale annual totals, and quantify the uncertainty associated with this upscaling. Here, we review some approaches to this problem and synthesize the work in the recent UK Greenhouse Gas Emissions and Feedbacks Program, which was designed to identify and address these challenges. Approaches to the scaling problem included: instrumentation developments which mean that near-continuous data sets can be produced with larger spatial coverage; geostatistical methods which address the problem of extrapolating to larger domains, using spatial information in the data; more rigorous statistical methods which characterize the uncertainty in extrapolating to longer time scales; analytical approaches to estimating model aggregation error; enhanced estimates of C flux measurement error; and novel uses of remote sensing data to calibrate process models for generating probabilistic regional C flux estimates.

41. Remotely Sensed Carbonyl Sulfide Constrains Model Estimates of Amazon Primary Productivity

Authors: James R. Stinecipher (University of California Merced, Lawrence Livermore National Laboratory); Philip Cameron-Smith (Lawrence Livermore National Laboratory); Le Kuai (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Norbert Glatthor (Karlsruhe Institute of Technology); Michael Höpfner (Karlsruhe Institute of Technology); Ian Baker (Colorado State University); Christian Beer (University of Hamburg); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Scot M. Miller (Johns Hopkins University); Nicholas Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); J. Elliott Campbell (University of California Santa Cruz)

Journal: Geophysical Research Letters

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2021gl096802

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Publication Date: 2022-04-22

Volume/Issue: Volume 49, Issue 9

References: 46

Cited by: 8 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: Understanding the magnitude of tropical gross primary production (GPP) is critical for carbon cycle modeling and climate projections, but this quantity is poorly constrained at regional scales. Biospheric uptake of carbonyl sulfide (OCS) provides a promising approach to estimating regional GPP. Here, we simulate OCS concentrations driven by surface flux scenarios encompassing a wide range of GPP estimates for the Amazon basin. We compare the model output to satellite retrievals and find a regional GPP estimate of $2375 \pm 914 \text{ g(C) m}^{-2} \text{ yr}^{-1}$, consistent with previous estimates, including the TRENDY model ensemble range of 1431-3812 $\text{g(C) m}^{-2} \text{ yr}^{-1}$.

42. CARDAMOM-FluxVal version 1.0: a FLUXNET-based validation system for CARDAMOM carbon and water flux estimates

Authors: Yan Yang; A. Anthony Bloom; Shuang Ma; Paul Levine; Alexander Norton; Nicholas C. Parazoo; John T. Reager; John Worden; Gregory R. Quetin; T. Luke Smallman; Mathew Williams; Liang Xu; Sassan Saatchi

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-1789-2022

URL: <https://doi.org/10.5194/gmd-15-1789-2022>

Publication Date: 2022-03-02

Volume/Issue: Volume 15, Issue 4

References: 48

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Jet Propulsion Laboratory

License: CC BY 4.0

Open Access: N/A

Abstract: . Land-atmosphere carbon and water exchanges have large uncertainty in terrestrial biosphere models (TBMs). Using observations to reduce TBM structural and parametric errors and uncertainty is a critical priority for both understanding and accurately predicting carbon and water fluxes. Recent implementations of the Bayesian CARbon Data-MODEl fraMework (CARDAMOM) have yielded key insights into ecosystem carbon and water cycling. CARDAMOM estimates parameters for an associated TBM of intermediate complexity (Data Assimilation Linked Ecosystem Carbon - DALEC). These CARDAMOM analyses - informed by co-located C and H₂O flux observations - have exhibited considerable skill in both representing the variability of assimilated observations and predicting withheld observations. CARDAMOM and DALEC have been continuously developed to accommodate new scientific challenges and an expanding variety of observational constraints. However, so far there has been no concerted effort to globally and systematically validate CARDAMOM performance across individual model-data fusion configurations. Here we use the FLUXNET 2015 dataset - an ensemble of 200+ eddy covariance flux tower sites - to formulate a concerted benchmarking framework for CARDAMOM carbon (photosynthesis and net C exchange) and water (evapotranspiration) flux estimates (CARDAMOM-FluxVal version 1.0). We present a concise set of skill metrics to evaluate CARDAMOM performance against both assimilated and withheld FLUXNET 2015 photosynthesis, net CO₂ exchange, and evapotranspiration estimates. We further demonstrate the potential for tailored CARDAMOM evaluations by categorizing performance in terms of (i) individual land-cover types, (ii) monthly, annual, and mean fluxes, and (iii) length of assimilation data. The CARDAMOM benchmarking system - along with the CARDAMOM driver files provided - can be readily repeated to support both the intercomparison between existing CARDAMOM model configurations and the formulation, development, and testing of new CARDAMOM model structures.

43. Spatiotemporal patterns and drivers of terrestrial dissolved organic carbon (DOC) leaching into the European river network

Authors: Céline Gommet; Ronny Lauerwald; Philippe Ciais; Bertrand Guenet; Haicheng Zhang; Pierre Regnier

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-13-393-2022

URL: <https://doi.org/10.5194/esd-13-393-2022>

Publication Date: 2022-02-21

Volume/Issue: Volume 13, Issue 1

References: 103

Cited by: 16 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Horizon 2020

License: CC BY 4.0

Open Access: N/A

Abstract: . Leaching of dissolved organic carbon (DOC) from soils into the river network is an important component of the land carbon (C) budget. At regional to global scales, its significance has been estimated through simple mass budgets, often using multi-year averages of observed fluvial DOC fluxes as a proxy of DOC leaching due to the limited availability of observations of the leaching flux itself. This procedure leads to a systematic underestimation of the leaching flux because of the decay of DOC during fluvial transport. Moreover, this procedure does not allow for revealing spatiotemporal variability in DOC leaching from soils, which is vital to better understand the drivers of DOC leaching and its impact on the local soil C budget. In this study, we use the land surface model (LSM) ORCHILEAK to simulate the terrestrial C budget, including leaching of DOC from the soil and its subsequent reactive transport through the river network of Europe. The model performance is evaluated not only against the sparse observations of the soil DOC leaching rate, but also against the more abundant observations of fluxes and reactivity of DOC in rivers, providing further evidence that our simulated DOC fluxes are realistic. The model is then used to simulate the spatiotemporal patterns of DOC leaching across Europe over the period 1972-2012, quantifying both the environmental drivers of these patterns and the impact of DOC leaching on the land C budget. Over the simulation period, we find that, on average, 14.3 Tg C yr⁻¹ of DOC is leached from land into European rivers, which is about 0.6 % of the terrestrial net primary production (NPP), a fraction significantly lower than that reported for tropical river networks. On average, 12.3 Tg C yr⁻¹ of the leached DOC is finally exported to the coast via the river network, and the rest is respired during transit. DOC leaching presents a large seasonal variability, with the maximum occurring in winter and the minimum in summer, except for most parts of northern Europe, where the maximum occurs in spring due to snowmelt. The DOC leaching rate is generally low in warm and dry regions, and high in the cold and wet regions of Europe. Furthermore, runoff and the ratio between runoff from shallower flow paths on one hand and deep drainage and groundwater flow on the other hand are the main drivers of the spatiotemporal variation of DOC leaching. Temperature, as a major control of DOC production and decomposition rates in the soils,

plays only a secondary role.

44. Definitions and methods to estimate regional land carbon fluxes for the second phase of the REgional Carbon Cycle Assessment and Processes Project (RECCAP-2)

Authors: Philippe Ciais; Ana Bastos; Frédéric Chevallier; Ronny Lauerwald; Ben Poulter; Josep G. Canadell; Gustaf Hugelius; Robert B. Jackson; Atul Jain; Matthew Jones; Masayuki Kondo; Ingrid T. Lujckx; Prabir K. Patra; Wouter Peters; Julia Pongratz; Ana Maria Roxana Petrescu; Shilong Piao; Chunjing Qiu; Celso Von Randow; Pierre Regnier; Marielle Saunois; Robert Scholes; Anatoly Shvidenko; Hanqin Tian; Hui Yang; Xuhui Wang; Bo Zheng

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-15-1289-2022

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Volume/Issue: Volume 15, Issue 3

References: 145

Cited by: 58 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Space Agency

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Open Access: N/A

Abstract: . Regional land carbon budgets provide insights into the spatial distribution of the land uptake of atmospheric carbon dioxide and can be used to evaluate carbon cycle models and to define baselines for land-based additional mitigation efforts. The scientific community has been involved in providing observation-based estimates of regional carbon budgets either by downscaling atmospheric CO₂ observations into surface fluxes with atmospheric inversions, by using inventories of carbon stock changes in terrestrial ecosystems, by upscaling local field observations such as flux towers with gridded climate and remote sensing fields, or by integrating data-driven or process-oriented terrestrial carbon cycle models. The first coordinated attempt to collect regional carbon budgets for nine regions covering the entire globe in the RECCAP-1 project has delivered estimates for the decade 2000–2009, but these budgets were not comparable between regions due to different definitions and component fluxes being reported or omitted. The recent recognition of lateral fluxes of carbon by human activities and rivers that connect CO₂ uptake in one area with its release in another also requires better definitions and protocols to reach harmonized regional budgets that can be summed up to a globe scale and compared with the atmospheric CO₂ growth rate and inversion results. In this study, using the international initiative RECCAP-2 coordinated by the Global Carbon Project, which aims to be an update to regional carbon budgets over the last 2 decades based on observations for 10 regions covering the globe with a better harmonization than the precursor project, we provide recommendations for using atmospheric inversion results to match bottom-up carbon accounting and models, and we define the different component fluxes of the net land atmosphere carbon exchange that should be reported by each research group in charge of each region. Special attention is given to lateral fluxes, inland water

fluxes, and land use fluxes.

45. Climate Sensitivities of Carbon Turnover Times in Soil and Vegetation: Understanding Their Effects on Forest Carbon Sequestration

Authors: Rong Ge (School of Government Audit Institute of Natural Resources and Environmental Audits Nanjing Audit University Nanjing China, Chinese Academy of Sciences); Honglin He (Chinese Academy of Sciences); Li Zhang (Chinese Academy of Sciences); Xiaoli Ren (Chinese Academy of Sciences); Mathew Williams (University of Edinburgh); Guirui Yu (Chinese Academy of Sciences); T. Luke Smallman (University of Edinburgh); Tao Zhou (Beijing Normal University); Pan Li (Institute of Surface-Earth System Science Tianjin University Tianjin China); Zongqiang Xie (Chinese Academy of Sciences); Silong Wang (Chinese Academy of Sciences); Huimin Wang (Chinese Academy of Sciences); Guoyi Zhou (Chinese Academy of Sciences); Qibin Zhang (Chinese Academy of Sciences); Anzhi Wang (Chinese Academy of Sciences); Zexin Fan (Chinese Academy of Sciences); Yiping Zhang (Chinese Academy of Sciences); Weijun Shen (Chinese Academy of Sciences); Huajun Yin (Chinese Academy of Sciences); Luxiang Lin (Chinese Academy of Sciences)

Journal: Journal of Geophysical Research: Biogeosciences

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2020jg005880

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Volume/Issue: Volume 127, Issue 3

References: 116

Cited by: 5 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: The high uncertainty associated with the response of terrestrial carbon (C) cycle to climate is dominated by ecosystem C turnover time (τ_{eco}). Although the relationship between τ_{eco} and climate has been extensively studied, significant knowledge gaps remain regarding the differential climate sensitivities of turnover time in major biomass (τ_{veg}) and soil (τ_{soil}) pools, and their effects on vegetation and soil C sequestration under climate change are poorly understood. Here, we collected multiple time series observations on soil and vegetation C from permanent plots in 10 Chinese forests and used model-data fusion to retrieve key C cycle process parameters that regulate τ_{soil} and τ_{veg} . Our analysis showed that τ_{veg} and τ_{soil} both decreased with increasing temperature and precipitation, and τ_{soil} was more than twice as sensitive (1.27 years/ degreesC, 1.70 years/100 mm) than τ_{veg} (0.53 years/ degreesC, 0.40 years/100 mm). The higher climate sensitivity of τ_{soil} caused a more rapid decrease in τ_{soil} than in τ_{veg} with increasing temperature and precipitation, thereby significantly reducing the difference between τ_{soil} and τ_{veg} (τ_{diff}) under warm and humid conditions. τ_{diff} , an indicator of the balance between the soil C input and exit rate, was strongly responsible for the variation (more than 50%) in soil C sequestration. Therefore, a smaller τ_{diff} under warm and humid conditions suggests a relatively lower contribution from soil C sequestration. This information has strong implications for understanding forest C-climate feedback,

predicting forest C sink distributions in soil and vegetation under climate change, and implementing C mitigation policies in forest plantations or soil conservation.

46. Resolving temperature limitation on spring productivity in an evergreen conifer forest using a model-data fusion framework

Authors: Stephanie G. Stettz; Nicholas C. Parazoo; A. Anthony Bloom; Peter D. Blanken; David R. Bowling; Sean P. Burns; Cédric Bacour; Fabienne Maignan; Brett Raczka; Alexander J. Norton; Ian Baker; Mathew Williams; Mingjie Shi; Yongguang Zhang; Bo Qiu

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-19-541-2022

URL: <https://doi.org/10.5194/bg-19-541-2022>

Publication Date: 2022-01-28

Volume/Issue: Volume 19, Issue 2

References: 84

Cited by: 6 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: U.S. Department of Energy, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . The flow of carbon through terrestrial ecosystems and the response to climate are critical but highly uncertain processes in the global carbon cycle. However, with a rapidly expanding array of in situ and satellite data, there is an opportunity to improve our mechanistic understanding of the carbon (C) cycle's response to land use and climate change. Uncertainty in temperature limitation on productivity poses a significant challenge to predicting the response of ecosystem carbon fluxes to a changing climate. Here we diagnose and quantitatively resolve environmental limitations on the growing-season onset of gross primary production (GPP) using nearly 2 decades of meteorological and C flux data (2000-2018) at a subalpine evergreen forest in Colorado, USA. We implement the CARbonData-MODEL fraMework (CARDAMOM) model-data fusion network to resolve the temperature sensitivity of spring GPP. To capture a GPP temperature limitation - a critical component of the integrated sensitivity of GPP to temperature - we introduced a cold-temperature scaling function in CARDAMOM to regulate photosynthetic productivity. We found that GPP was gradually inhibited at temperatures below 6.0 °C (± 2.6 °C) and completely inhibited below -7.1 °C (± 1.1 °C). The addition of this scaling factor improved the model's ability to replicate spring GPP at interannual and decadal timescales ($r=0.88$), relative to the nominal CARDAMOM configuration ($r=0.47$), and improved spring GPP model predictability outside of the data assimilation training period ($r=0.88$). While cold-temperature limitation has an important influence on spring GPP, it does not have a significant impact on integrated growing-season GPP, revealing that other environmental controls, such as precipitation, play a more important role in annual productivity. This study highlights growing-season onset temperatures as a key limiting factor for spring growth in winter-dormant evergreen forests, which is critical in understanding future responses to climate change.

47. Four years of global carbon cycle observed from the Orbiting Carbon Observatory 2 (OCO-2) version 9 and in situ data and comparison to OCO-2 version 7

Authors: H      Peiro; Sean Crowell; Andrew Schuh; David F. Baker; Chris O'Dell; Andrew R. Jacobson; Fr        Chevallier; Junjie Liu; Annmarie Eldering; David Crisp; Feng Deng; Brad Weir; Sourish Basu; Matthew S. Johnson; Sajeev Philip; Ian Baker

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-22-1097-2022

URL: <https://doi.org/10.5194/acp-22-1097-2022>

Publication Date: 2022-01-24

Volume/Issue: Volume 22, Issue 2

References: 130

Cited by: 67 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: European Commission, National Aeronautics and Space Administration

License: CC BY 4.0

Open Access: N/A

Abstract: . The Orbiting Carbon Observatory 2 (OCO-2) satellite has been providing information to estimate carbon dioxide (CO₂) fluxes at global and regional scales since 2014 through the combination of CO₂ retrievals with top-down atmospheric inversion methods. Column average CO₂ dry-air mole fraction retrievals have been constantly improved. A bias correction has been applied in the OCO-2 version 9 retrievals compared to the previous OCO-2 version 7r improving data accuracy and coverage. We study an ensemble of 10 atmospheric inversions all characterized by different transport models, data assimilation algorithms, and prior fluxes using first OCO-2 v7 in 2015-2016 and then OCO-2 version 9 land observations for the longer period 2015-2018. Inversions assimilating in situ (IS) measurements have also been used to provide a baseline against which the satellite-driven results are compared. The time series at different scales (going from global to regional scales) of the models emissions are analyzed and compared to each experiment using either OCO-2 or IS data. We then evaluate the inversion ensemble based on the dataset from the Total Carbon Column Observing Network (TCCON), aircraft, and in situ observations, all independent from assimilated data. While we find a similar constraint of global total carbon emissions between the ensemble spread using IS and both OCO-2 retrievals, differences between the two retrieval versions appear over regional scales and particularly in tropical Africa. A difference in the carbon budget between v7 and v9 is found over this region, which seems to show the impact of corrections applied in retrievals. However, the lack of data in the tropics limits our conclusions, and the estimation of carbon emissions over tropical Africa require further analysis.

48. Comments on Global Carbon Budget 2021 (essd-2021-386)

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/essd-2021-386-cc1

URL: <https://doi.org/10.5194/essd-2021-386-cc1>

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Carbon sink|Global Change|Earth system science

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

49. Variability of ecosystem carbon source from microbial respiration is controlled by rainfall dynamics

Authors: Heng Huang (Department of Biological and Agricultural Engineering, Texas A&M; University, College Station, TX 77843;); Salvatore Calabrese (Department of Biological and Agricultural Engineering, Texas A&M; University, College Station, TX 77843;); Ignacio Rodriguez-Iturbe (Department of Civil and Environmental Engineering, Texas A&M; University, College Station, TX 77843, Department of Biological and Agricultural Engineering, Texas A&M; University, College Station, TX 77843;., Department of Ocean Engineering, Texas A&M; University, College Station, TX 77843;)

Journal: Proceedings of the National Academy of Sciences

Publisher: Proceedings of the National Academy of Sciences

DOI: 10.1073/pnas.2115283118

URL: <https://doi.org/10.1073/pnas.2115283118>

Publication Date: 2021-12-20

Volume/Issue: Volume 118, Issue 52

References: 50

Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: USDA | National Institute of Food and Agriculture

License: N/A

Open Access: N/A

Abstract: Significance The variability of the different ecosystem carbon fluxes directly influences the global carbon budget and future climate trajectory. Heterotrophic respiration is particularly uncertain and the extent to which its dynamics are driven by climate, soil, and vegetation properties remains poorly understood. Here we combine a global carbon flux dataset with a probabilistic model of microbial growth to show that despite its complexity the ecosystem-scale heterotrophic respiration can be described as a function of rainfall characteristics and vegetation primary productivity, regardless of ecosystem type. This emergent simplicity in ecosystem-scale heterotrophic respiration may help close the ecosystem carbon budgets and provide a quantitative framework to forecast the possible impacts of climate change on the soil carbon balance.

50. Parameter uncertainty dominates C-cycle forecast errors over most of Brazil for the 21st century

Authors: Thomas Luke Smallman; David Thomas Milodowski; Eráclito Sousa Neto; Gerbrand Koren; Jean Ometto; Mathew Williams

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-12-1191-2021

URL: <https://doi.org/10.5194/esd-12-1191-2021>

Publication Date: 2021-11-23

Volume/Issue: Volume 12, Issue 4

References: 114

Cited by: 13 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: UK Space Agency, National Centre for Earth Observation, Newton Fund (and 1 more)

License: CC BY 4.0

Open Access: N/A

Abstract: . Identification of terrestrial carbon (C) sources and sinks is critical for understanding the Earth system as well as mitigating and adapting to climate change resulting from greenhouse gas emissions. Predicting whether a given location will act as a C source or sink using terrestrial ecosystem models (TEMs) is challenging due to net flux being the difference between far larger, spatially and temporally variable fluxes with large uncertainties. Uncertainty in projections of future dynamics, critical for policy evaluation, has been determined using multi-TEM intercomparisons, for various emissions scenarios. This approach quantifies structural and forcing errors. However, the role of parameter error within models has not been determined. TEMs typically have defined parameters for specific plant functional types generated from the literature. To ascertain the importance of parameter error in forecasts, we present a Bayesian analysis that uses data on historical and current C cycling for Brazil to parameterise five TEMs of varied complexity with a retrieval of model error covariance at 1° spatial resolution. After evaluation against data from 2001-2017, the parameterised models are simulated to 2100 under four climate change scenarios spanning the likely range of climate projections. Using multiple models, each with per pixel parameter ensembles, we partition forecast uncertainties. Parameter uncertainty dominates across most of Brazil when simulating future stock changes in biomass C and dead organic matter (DOM). Uncertainty of simulated biomass change is most strongly correlated with net primary productivity allocation to wood (NPP_{wood}) and mean residence time of wood (MRT_{wood}). Uncertainty of simulated DOM change is most strongly correlated with MRT_{soil} and NPP_{wood}. Due to the coupling between these variables and C stock dynamics being bi-directional, we argue that using repeated estimates of woody biomass will provide a valuable constraint needed to refine predictions of the future carbon cycle. Finally, evaluation of our multi-model analysis shows that wood litter contributes substantially to fire emissions, necessitating a greater understanding of wood litter C cycling than is typically considered in large-scale TEMs.

51. Covariation of Airborne Biogenic Tracers (CO₂, COS, and CO) Supports Stronger Than Expected Growing Season Photosynthetic Uptake in the Southeastern US

Authors: Nicholas C. Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Bianca C. Baier (NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA, Cooperative Institute for Research in Environmental Sciences University of Colorado-Boulder Boulder CO USA); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Meemong Lee (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Le Kuai (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Yoichi Shiga (Universities Space Research Association); Ian Baker (Colorado State University); Mary E. Whelan (Department of Environmental Sciences Rutgers, The State University of New Jersey New Brunswick NJ USA); Sha Feng (Pennsylvania State University); Maarten Krol (Wageningen University, Utrecht University); Colm Sweeney (NOAA Earth System Research Laboratory Global Monitoring Division Boulder CO USA); Benjamin R. Runkle (University of Arkansas); Elahe Tajfar (University of Arkansas); Kenneth J. Davis (Pennsylvania State University)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

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URL: <https://doi.org/10.1029/2021gb006956>

Publication Date: 2021-09-12

Volume/Issue: Volume 35, Issue 10

References: 98

Cited by: 7 publications

Language: en

Subjects: N/A

Keywords: carbon cycle, carbonyl sulfide, act-America, ecosystem, carbon monoxide (and 1 more)

Funding: NASA

License: N/A

Open Access: N/A

Abstract: The Atmospheric Carbon Transport (ACT)-America Earth Venture mission conducted five airborne campaigns across four seasons from 2016 to 2019, to study the transport and fluxes of Greenhouse gases across the eastern United States. Unprecedented spatial sampling of atmospheric tracers (CO₂, carbon monoxide [CO], carbonyl sulfide [COS]) related to biospheric processes offers opportunities to improve our qualitative and quantitative understanding of seasonal and spatial patterns of biospheric carbon uptake. Here, we examine co-variation of boundary layer enhancements of CO₂, CO, and COS across three diverse regions: the crop-dominated Midwest, evergreen-dominated South, and deciduous broadleaf-dominated Northeast. To understand the biogeochemical processes controlling these tracers, we compare the observed co-variation to simulated co-variation resulting from model- and satellite- constrained surface carbon fluxes. We found indication of a common terrestrial

biogenic sink of CO₂ and COS and secondary production of CO from biogenic sources in summer throughout the eastern US, driven by stomatal conductance. Upper Midwest crops drive CO₂ and COS depletion from early to late summer. Northeastern temperate forests drive CO₂ and COS depletion in late summer. The unprecedented ACT-America flask samples uncovered evidence that southern humid temperate forests photosynthesize and absorb CO₂ and COS, and emit CO precursors, deep into the growing season. Satellite- constrained carbon fluxes capture much of the observed seasonal and spatial variability, but underestimate the magnitude of net CO₂ and COS depletion in the South, indicating a stronger than expected net sink of CO₂ in late summer. Additional sampling of the South will more accurately constrain underlying biological processes and climate sensitivities governing southern carbon dynamics.

52. Review of Yang et al. (gmd-2021-190)

Authors: No authors listed

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-2021-190-rc1

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: FluxNet

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

53. Evaluating two land surface models for Brazil using a full carbon cycle benchmark with uncertainties

Authors: Auguste Caen (University of Edinburgh); T. Luke Smallman (University of Edinburgh); Aline Anderson de Castro (Impacts, Adaptation and Vulnerability Division INPE São José dos Campos SP Brazil); Eddy Robertson (Met Office Hadley Centre, Met Office); Celso von Randow (Impacts, Adaptation and Vulnerability Division INPE São José dos Campos SP Brazil); Manoel Cardoso (Impacts, Adaptation and Vulnerability Division INPE São José dos Campos SP Brazil); Mathew Williams (University of Edinburgh)

Journal: Climate Resilience and Sustainability

Publisher: Wiley

DOI: 10.1002/cli2.10

URL: <https://doi.org/10.1002/cli2.10>

Publication Date: 2021-08-13

Volume/Issue: Volume 1, Issue 1

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Open Access: N/A

Abstract: Forecasts of tropical ecosystem C cycling diverge among models due to differences in simulation of internal processes such as turnover, or transit times, of carbon pools. Estimates of these processes for the recent past are needed to test model representations, and so build confidence in model forecasts within and across biomes. Here, we evaluate carbon cycle process representation in two land surface models [Joint UK Land Environment Simulator (JULES) and Integrated Model of Land Surface Processes (INLAND)] for the period 2001-10 across Brazilian biomes. Model outputs are evaluated using the ILAMB system. Probabilistic benchmarking data were created using the carbon data model framework that assimilates observational times series of leaf area index and maps of woody biomass and soil C. New custom uncertainty metrics assess if models are within benchmark uncertainties. Simulations are better in homogeneous areas of vegetation type, and are less robust at ecotones between biomes, likely due to disturbance effects and parameter errors. Gross biosphere-atmosphere fluxes are robustly modelled across Brazil. However, benchmark uncertainty is too high on net ecosystem exchange to provide an accurate evaluation of the models. The LSMs have significant differences in internal carbon allocation and the dynamics of the different C pools. JULES models dead C stocks more accurately while living C stocks are best resolved for INLAND. JULES' over-estimate of the C wood pool results from over-estimation of both inputs to wood and the transit time of wood. INLAND's under-estimate of dead C stocks arises from an under-estimate of the transit time of dead organic matter. The models are better at simulating annual averages than seasonal variation of fluxes. Analyses of monthly net C exchanges show that INLAND correctly simulates seasonality, but over-estimates amplitudes, whereas JULES correctly simulates the annual amplitudes,

but is out of phase with the benchmark.

54. Reanalysis in Earth System Science: Toward Terrestrial Ecosystem

Reanalysis

Authors: R. Baatz (International Soil Modelling Consortium, Forschungszentrum Jülich); H. J. Hendricks Franssen (Forschungszentrum Jülich); E. Euskirchen (University of Alaska Fairbanks); D. Sihi (Emory University); M. Dietze (Boston University); S. Ciavatta (Plymouth Marine Laboratory); K. Fennel (Dalhousie University); H. Beck (Princeton University); G. De Lannoy (KU Leuven); V. R. N. Pauwels (Monash University); A. Raiho (Colorado State University); C. Montzka (Forschungszentrum Jülich); M. Williams (University of Edinburgh); U. Mishra (Sandia National Laboratory); C. Poppe (Forschungszentrum Jülich); S. Zacharias (Department of Monitoring and Exploration Technologies UFZ Helmholtz Centre for Environmental Research Leipzig Germany); A. Lausch (Department Computational Landscape Ecology Helmholtz Centre for Environmental Research-UFZ Leipzig Germany, Geography Department Humboldt University Berlin Berlin Germany); L. Samaniego (Department Computational Hydrosystems Helmholtz Centre for Environmental Research - UFZ Leipzig Germany); K. Van Looy (OVAM, International Policy Unit Mechelen Belgium); H. Bogaen (Forschungszentrum Jülich); M. Adamescu (University of Bucharest); M. Mirtl (Department of Monitoring and Exploration Technologies UFZ Helmholtz Centre for Environmental Research Leipzig Germany); A. Fox (Joint Center for Satellite Data Assimilation UCAR Boulder CO USA); K. Goergen (Centre for High-Performance Scientific Computing in Terrestrial Systems Geoverbund ABC/J Jülich Germany, Forschungszentrum Jülich); B. S. Naz (Centre for High-Performance Scientific Computing in Terrestrial Systems Geoverbund ABC/J Jülich Germany, Forschungszentrum Jülich); Y. Zeng (Faculty of Geo-information Science and Earth Observation (ITC) University of Twente Enschede The Netherlands); H. Vereecken (International Soil Modelling Consortium, Forschungszentrum Jülich)

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Abstract: A reanalysis is a physically consistent set of optimally merged simulated model states and historical observational data, using data assimilation. High computational costs for modeled processes and assimilation algorithms has led to Earth system specific reanalysis products for the atmosphere, the ocean and the land separately. Recent developments include the advanced uncertainty quantification and the generation of biogeochemical reanalysis for land and ocean. Here, we review atmospheric and oceanic reanalyses, and more in detail biogeochemical ocean and terrestrial reanalyses. In particular, we identify land surface, hydrologic and carbon cycle reanalyses which are

nowadays produced in targeted projects for very specific purposes. Although a future joint reanalysis of land surface, hydrologic, and carbon processes represents an analysis of important ecosystem variables, biotic ecosystem variables are assimilated only to a very limited extent. Continuous data sets of ecosystem variables are needed to explore biotic-abiotic interactions and the response of ecosystems to global change. Based on the review of existing achievements, we identify five major steps required to develop terrestrial ecosystem reanalysis to deliver continuous data streams on ecosystem dynamics.

55. CARDAMOM-FluxVal Version 1.0: a FLUXNET-based Validation System for CARDAMOM Carbon and Water Flux Estimates

Authors: Yan Yang; A. Anthony Bloom; Shuang Ma; Paul Levine; Alexander Norton; Nicholas C. Parazoo; John T. Reager; John Worden; Gregory R. Quetin; T. Luke Smallman; Mathew Williams; Liang Xu; Sassan Saatchi

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Keywords: FluxNet|Benchmarking|Land Cover

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Abstract: . Land-atmosphere carbon and water exchanges have large uncertainty in land surface and biosphere models. Using observations to reduce land biosphere model structural and parametric errors is a key priority for both understanding and accurately predicting carbon and water fluxes. Recent implementations of the Bayesian CARDAMOM model-data fusion framework have yielded key insights into ecosystem carbon and water cycling. CARDAMOM analyses--informed by co-located C and H₂O flux observations--have exhibited considerable skill in both representing the variability of assimilated observations and predicting withheld observations. While CARDAMOM model configurations (namely CARDAMOM-compatible biogeochemical model structures) have been continuously developed to accommodate new scientific challenges and an expanding variety of observational constraints, there has so far been no concerted effort to globally and systematically validate CARDAMOM performance across individual model-data fusion configurations. Here we use the FLUXNET-2015 dataset--an ensemble of 200+ eddy covariance flux tower sites--to formulate a concerted benchmarking framework for CARDAMOM carbon (GPP, NEE) and water (ET) flux estimates (CARDAMOM-FLUXVal version 1.0). We present a concise set of skill metrics to evaluate CARDAMOM performance against both assimilated and withheld FLUXNET-2015 GPP, NEE and ET data. We further demonstrate the potential for tailored CARDAMOM evaluations by categorizing performance in terms of (i) individual land cover types, (ii) monthly, annual and mean fluxes, and (iii) length of assimilation data. The CARDAMOM benchmarking system--along with CARDAMOM driver files provided--can be readily repeated to support both the intercomparison between existing CARDAMOM model configurations and the formulation, development and testing of new CARDAMOM model structures.

56. RC Comment on bg-2021-144

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Journal: Unknown Journal

Publisher: Copernicus GmbH

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Language: N/A

Subjects: N/A

Keywords: Grassland ecosystem|Biogeochemical Cycle

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Abstract: N/A

57. Spatio-temporal patterns and drivers of terrestrial Dissolved Organic Carbon (DOC) leaching to the European river network

Authors: Céline Annick Sylviane Gomet; Ronny Lauerwald; Philippe Ciais; Bertrand Guenet; Haicheng Zhang; Pierre Regnier

Journal: Unknown Journal

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Keywords: N/A

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Open Access: N/A

Abstract: . Leaching of dissolved organic carbon (DOC) from soils to the river network is an important component of the land carbon (C) budget. At regional to global scales, its significance has been estimated through simple mass budgets, often using multi-year averages of observed fluvial DOC fluxes as proxy of DOC leaching due to the limited availability of observations of the leaching flux itself. This procedure leads to a systematic underestimation of the leaching flux because of the reactivity of DOC during fluvial transport. Moreover, this procedure does not allow to reveal spatio-temporal variability in DOC leaching from soils, which is needed to better understand the drivers of DOC leaching and its impact on the local soil C budget. In this study, we use the land surface model ORCHILEAK to simulate the terrestrial C budget including leaching of DOC from the soil and its subsequent reactive transport through the river network of Europe. The model performance is not only evaluated against the sparse observations of DOC leaching, but also against the more abundant observations of fluxes and reactivity of DOC in rivers, providing further evidence that our simulated DOC leaching fluxes are realistic. The model is then used to simulate the spatio-temporal patterns of DOC leaching across Europe over the period 1972 to 2012, quantifying both the environmental drivers of these patterns as well as the impact of DOC leaching on the land C budget. Over the simulation period, we find that on average 14.3 TgC yr⁻¹ of DOC is leached from land to European rivers, which is only about 0.6 % of the terrestrial net primary production, a fraction about one order of magnitude lower than reported for tropical river networks. Of the DOC leaching, on average 12.3 TgC yr⁻¹ is exported to the coast via the river network, the rest being respired in transit. DOC leaching presents a large seasonal variability, with a maximum occurring in winter and a minimum in summer, except for the Northern most part of Europe where the maximum occurs in spring due to the snow melt. DOC leaching rate is generally lower in warm and dry regions, and higher in cold and wet regions of Europe. Furthermore, runoff, and the ratio between runoff from shallower flow paths vs. deep drainage and groundwater flow, is the main driver of the spatial variation of DOC leaching. Temperature, as a major control of DOC production and decomposition rates in the soils, plays only a secondary role.

58. Resolving temperature limitation on spring productivity in an evergreen conifer forest using a model-data fusion framework

Authors: Stephanie G. Stettz; Nicholas C. Parazoo; A. Anthony Bloom; Peter D. Blanken; David R. Bowling; Sean P. Burns; Cédric Bacour; Fabienne Maignan; Brett Raczka; Alexander J. Norton; Ian Baker; Mathew Williams; Mingjie Shi; Yongguang Zhang; Bo Qiu

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Keywords: Predictability

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Open Access: N/A

Abstract: . The flow of carbon through terrestrial ecosystems and the response to climate is a critical but highly uncertain process in the global carbon cycle. However, with a rapidly expanding array of in situ and satellite data, there is an opportunity to improve our mechanistic understanding of the carbon (C) cycle's response to land use and climate change. Uncertainty in temperature limitation on productivity pose a significant challenge to predicting the response of ecosystem carbon fluxes to a changing climate. Here we diagnose and quantitatively resolve environmental limitations on growing season onset of gross primary production (GPP) using nearly two decades of meteorological and C flux data (2000-2018) at a subalpine evergreen forest in Colorado USA. We implement the CARDAMOM model-data fusion network to resolve the temperature sensitivity of spring GPP. To capture a GPP temperature limitation - a critical component of integrated sensitivity of GPP to temperature - we introduced a cold temperature scaling function in CARDAMOM to regulate photosynthetic productivity. We found that GPP was gradually inhibited at temperature below 6.0 degreesC (± 2.6 degreesC) and completely inhibited below -7.1 degreesC (± 1.1 degreesC). The addition of this scaling factor improved the model's ability to replicate spring GPP at interannual and decadal time scales ($r = 0.88$), relative to the nominal CARDAMOM configuration ($r = 0.47$), and improved spring GPP model predictability outside of the data assimilation training period ($r = 0.88$). While cold temperature limitation has an important influence on spring GPP, it does not have a significant impact on integrated growing season GPP, revealing that other environmental controls, such as precipitation, play a more important role in annual productivity. This study highlights growing season onset temperature as a key limiting factor for spring growth in winter-dormant evergreen forests, which is critical in understanding future responses to climate change.

59. Inferring management and predicting sub-field scale C dynamics in UK grasslands using biogeochemical modelling and satellite-derived leaf area data

Authors: Vasileios Myrgiotis; Paul Harris; Andrew Reville; Hadewij Sint; Mathew Williams

Journal: Agricultural and Forest Meteorology

Publisher: Elsevier BV

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Abstract: N/A

60. Optimal model complexity for terrestrial carbon cycle prediction

Authors: Caroline A. Famiglietti; T. Luke Smallman; Paul A. Levine; Sophie Flack-Prain; Gregory R. Quetin; Victoria Meyer; Nicholas C. Parazoo; Stephanie G. Stettz; Yan Yang; Damien Bonal; A. Anthony Bloom; Mathew Williams; Alexandra G. Konings

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Abstract: . The terrestrial carbon cycle plays a critical role in modulating the interactions of climate with the Earth system, but different models often make vastly different predictions of its behavior. Efforts to reduce model uncertainty have commonly focused on model structure, namely by introducing additional processes and increasing structural complexity. However, the extent to which increased structural complexity can directly improve predictive skill is unclear. While adding processes may improve realism, the resulting models are often encumbered by a greater number of poorly determined or over-generalized parameters. To guide efficient model development, here we map the theoretical relationship between model complexity and predictive skill. To do so, we developed 16 structurally distinct carbon cycle models spanning an axis of complexity and incorporated them into a model-data fusion system. We calibrated each model at six globally distributed eddy covariance sites with long observation time series and under 42 data scenarios that resulted in different degrees of parameter uncertainty. For each combination of site, data scenario, and model, we then predicted net ecosystem exchange (NEE) and leaf area index (LAI) for validation against independent local site data. Though the maximum model complexity we evaluated is lower than most traditional terrestrial biosphere models, the complexity range we explored provides universal insight into the inter-relationship between structural uncertainty, parametric uncertainty, and model forecast skill. Specifically, increased complexity only improves forecast skill if parameters are adequately informed (e.g., when NEE observations are used for calibration). Otherwise, increased complexity can degrade skill and an intermediate-complexity model is optimal. This finding remains consistent regardless of whether NEE or LAI is predicted. Our COMPLEXityEXperiment (COMPLEX) highlights the importance of robust observation-based parameterization for land surface modeling and suggests that data characterizing net carbon fluxes will be key to improving decadal predictions of high-dimensional terrestrial biosphere models.

61. Combining Process Modelling and LAI Observations to Diagnose Winter Wheat Nitrogen Status and Forecast Yield

Authors: Andrew Revill (University of Edinburgh); Vasileios Myrghiotis (University of Edinburgh); Anna Florence (Agriculture, Horticulture and Engineering Science Department, Scotland's Rural College, Edinburgh EH9 3JG, UK); Stephen Hoad (Agriculture, Horticulture and Engineering Science Department, Scotland's Rural College, Edinburgh EH9 3JG, UK); Robert Rees (Agriculture, Horticulture and Engineering Science Department, Scotland's Rural College, Edinburgh EH9 3JG, UK); Alasdair MacArthur (University of Edinburgh); Mathew Williams (University of Edinburgh)

Journal: Agronomy

Publisher: MDPI AG

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Funding: Biotechnology and Biological Sciences Research Council, Natural Environment Research Council

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Abstract: Climate, nitrogen (N) and leaf area index (LAI) are key determinants of crop yield. N additions can enhance yield but must be managed efficiently to reduce pollution. Complex process models estimate N status by simulating soil-crop N interactions, but such models require extensive inputs that are seldom available. Through model-data fusion (MDF), we combine climate and LAI time-series with an intermediate-complexity model to infer leaf N and yield. The DALEC-Crop model was calibrated for wheat leaf N and yields across field experiments covering N applications ranging from 0 to 200 kg N ha⁻¹ in Scotland, UK. Requiring daily meteorological inputs, this model simulates crop C cycle responses to LAI, N and climate. The model, which includes a leaf N-dilution function, was calibrated across N treatments based on LAI observations, and tested at validation plots. We showed that a single parameterization varying only in leaf N could simulate LAI development and yield across all treatments--the mean normalized root-mean-square-error (NRMSE) for yield was 10%. Leaf N was accurately retrieved by the model (NRMSE = 6%). Yield could also be reasonably estimated (NRMSE = 14%) if LAI data are available for assimilation during periods of typical N application (April and May). Our MDF approach generated robust leaf N content estimates and timely yield predictions that could complement existing agricultural technologies. Moreover, EO-derived LAI products at high spatial and temporal resolutions provides a means to apply our approach regionally. Testing yield predictions from this approach over agricultural fields is a critical next step to determine broader utility.

62. Carbon Monitoring System Flux Net Biosphere Exchange 2020 (CMS-Flux NBE 2020)

Authors: Junjie Liu; Latha Baskaran; Kevin Bowman; David Schimel; A. Anthony Bloom; Nicholas C. Parazoo; Tomohiro Oda; Dustin Carroll; Dimitris Menemenlis; Joanna Joiner; Roisin Commane; Bruce Daube; Lucianna V. Gatti; Kathryn McKain; John Miller; Britton B. Stephens; Colm Sweeney; Steven Wofsy

Journal: Earth System Science Data

Publisher: Copernicus GmbH

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Abstract: . Here we present a global and regionally resolved terrestrial netbiosphere exchange (NBE) dataset with corresponding uncertainties between 2010–2018: Carbon Monitoring System Flux Net Biosphere Exchange 2020 (CMS-Flux NBE 2020). It is estimated using the NASA Carbon Monitoring System Flux (CMS-Flux) top-down flux inversion system that assimilates column CO₂ observations from the Greenhouse Gases Observing Satellite (GOSAT) and NASA's Observing Carbon Observatory 2 (OCO-2). The regional monthly fluxes are readily accessible as tabular files, and the gridded fluxes are available in NetCDF format. The fluxes and their uncertainties are evaluated by extensively comparing the posterior CO₂ mole fractions with CO₂ observations from aircraft and the NOAA marine boundary layer reference sites. We describe the characteristics of the dataset as the global total, regional climatological mean, and regional annual fluxes and seasonal cycles. We find that the global total fluxes of the dataset agree with atmospheric CO₂ growth observed by the surface-observation network within uncertainty. Averaged between 2010 and 2018, the tropical regions range from close to neutral in tropical South America to a net source in Africa; these contrast with the extra-tropics, which are a net sink of 2.5±0.3 Gt C/year. The regional satellite-constrained NBE estimates provide a unique perspective for understanding the terrestrial biosphere carbon dynamics and monitoring changes in regional contributions to the changes of atmospheric CO₂ growth rate. The gridded and regional aggregated dataset can be accessed at <https://doi.org/10.25966/4v02-c391> (Liu et al., 2020).

63. Mathematical modelling of collagen fibres rearrangement during the tendon healing process

Authors: José Antonio Carrillo; Martin Parisot; Zuzanna Szymańska

Journal: Kinetic & Related Models

Publisher: American Institute of Mathematical Sciences (AIMS)

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Subjects: N/A

Keywords: Kinetic model, Alignment, Integrodifferential equations, Tendon healing, Collagen remodelling

Funding: N/A

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Open Access: N/A

Abstract: N/A

64. Aircraft-based inversions quantify the importance of wetlands and livestock for Upper Midwest methane emissions

Authors: Xueying Yu; Dylan B. Millet; Kelley C. Wells; Daven K. Henze; Hansen Cao; Timothy J. Griffiths; Eric A. Kort; Genevieve Plant; Malte J. Deventer; Randall K. Kolka; D. Tyler Roman; Kenneth J. Davis; Ankur R. Desai; Bianca C. Baier; Kathryn McKain; Alan C. Czarnetzki; A. Anthony Bloom

Journal: Atmospheric Chemistry and Physics

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Funding: National Aeronautics and Space Administration, National Science Foundation, U.S. Department of Energy

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Abstract: . We apply airborne measurements across three seasons (summer, winter and spring 2017-2018) in a multi-inversion framework to quantify methane emissions from the US Corn Belt and Upper Midwest, a key agricultural and wetland source region. Combining our seasonal results with prior fall values we find that wetlands are the largest regional methane source (32 %, 20 [16-23] Gg/d), while livestock (enteric/manure; 25 %, 15 [14-17] Gg/d) are the largest anthropogenic source. Natural gas/petroleum, waste/landfills, and coal mines collectively make up the remainder. Optimized fluxes improve model agreement with independent datasets within and beyond the study timeframe. Inversions reveal coherent and seasonally dependent spatial errors in the WetCHARTs ensemble mean wetland emissions, with an underestimate for the Prairie Pothole region but an overestimate for Great Lakes coastal wetlands. Wetland extent and emission temperature dependence have the largest influence on prediction accuracy; better representation of coupled soil temperature-hydrology effects is therefore needed. Our optimized regional livestock emissions agree well with the Gridded EPA estimates during spring (to within 7 %) but are ~ 25 % higher during summer and winter. Spatial analysis further shows good top-down and bottom-up agreement for beef facilities (with mainly enteric emissions) but larger (~ 30 %) seasonal discrepancies for dairies and hog farms (with > 40 % manure emissions). Findings thus support bottom-up enteric emission estimates but suggest errors for manure; we propose that the latter reflects inadequate treatment of management factors including field application. Overall, our results confirm the importance of intensive animal agriculture for regional methane emissions, implying substantial mitigation opportunities through improved management.

65. Satellite Observations of the Tropical Terrestrial Carbon Balance and Interactions With the Water Cycle During the 21st Century

Authors: John Worden (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Sassan Saatchi (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Michael Keller (EMBRAPA Agricultural Informatics Campinas Brazil, National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, U.S. Department of Agriculture Forest Service International Institute of Tropical Forestry Campinas Brazil); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Junjie Liu (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Nicholas Parazoo (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Joshua B. Fisher (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Kevin Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, University of California Los Angeles); John T. Reager (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Kristen Fahy (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); David Schimel (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory); Rong Fu (University of California Los Angeles); Sarah Worden (University of California Los Angeles); Yi Yin (California Institute of Technology Geological and Planetary Science Pasadena CA USA); Pierre Gentile (Columbia University); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA); Gregory R. Quetin (Department of Earth System Science Stanford University Stanford CA USA); Mathew Williams (University of Edinburgh); Helen Worden (National Center for Atmospheric Research); Mingjie Shi (University of California Los Angeles, Pacific Northwest National Labs); Armineh Barkhordarian (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory, University of Hamburg)

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Abstract: A constellation of satellites is now in orbit providing information about terrestrial carbon and water storage and fluxes. These combined observations show that the tropical biosphere has changed significantly in the last 2 decades from the combined effects of climate variability and land use. Large areas of forest have been cleared in both wet and dry forests, increasing the source of carbon to the atmosphere. Concomitantly, tropical fire emissions have declined, at least until 2016, from changes in land-use practices and rainfall, increasing the net carbon sink. Measurements of carbon stocks and

fluxes from disturbance and recovery and of vegetation photosynthesis show significant regional variability of net biosphere exchange and gross primary productivity across the tropics and are tied to seasonal and interannual changes in water fluxes and storage. Comparison of satellite based estimates of evapotranspiration, photosynthesis, and the deuterium content of water vapor with patterns of total water storage and rainfall demonstrate the presence of vegetation-atmosphere interactions and feedback mechanisms across tropical forests. However, these observations of stocks, fluxes and inferred interactions between them do not point unambiguously to either positive or negative feedbacks in carbon and water exchanges. These ambiguities highlight the need for assimilation of these new measurements with Earth System models for a consistent assessment of process interactions, along with focused field campaigns that integrate ground, aircraft and satellite measurements, to quantify the controlling carbon and water processes and their feedback mechanisms.

66. Using satellite data to identify the methane emission controls of South Sudan's wetlands

Authors: Sudhanshu Pandey; Sander Houweling; Alba Lorente; Tobias Borsdorff; Maria Tsivlidou; A. Anthony Bloom; Benjamin Poulter; Zhen Zhang; Ilse Aben

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Open Access: N/A

Abstract: . The TROPOspheric Monitoring Instrument (TROPOMI) provides observations of atmospheric methane (CH₄) at an unprecedented combination of high spatial resolution and daily global coverage. Hu et al. (2018) reported unexpectedly large methane enhancements over South Sudan in these observations. Here we assess methane emissions from the wetlands of South Sudan using 2 years (December 2017–November 2019) of TROPOMI total column methane observations. We estimate annual wetland emissions of 7.4 ± 3.2 Tg yr⁻¹, which agrees with the multiyear GOSAT inversion of Lunt et al. (2019) but is an order of magnitude larger than estimates from wetland process models. This disagreement may be explained by the underestimation (by up to 4 times) of inundation extent by the hydrological schemes used in those models. We investigate the seasonal cycle of the emissions and find the lowest emissions during the June–August season when the process models show the largest emissions. Using satellite–altimetry-based river water height measurements, we infer that this seasonal mismatch is likely due to a seasonal mismatch in inundation extent. In models, inundation extent is controlled by regional precipitation scaled to static wetland extent maps, whereas the actual inundation extent is driven by water inflow from rivers like the White Nile and the Sobat. We find the lowest emissions in the highest precipitation and lowest temperature season (June–August, JJA) when models estimate large emissions. In general, our emission estimates show better agreement in terms of both seasonal cycle and annual mean with model estimates that use a stronger temperature dependence. This suggests that temperature might be a stronger control for the South Sudan wetland emissions than currently assumed by models. Our findings demonstrate the use of satellite instruments for quantifying emissions from inaccessible and uncertain tropical wetlands, providing clues for the improvement of process models and thereby improving our understanding of the currently uncertain contribution of wetlands to the global methane budget.

67. Lagged effects regulate the inter-annual variability of the tropical carbon balance

Authors: A. Anthony Bloom; Kevin W. Bowman; Junjie Liu; Alexandra G. Konings; John R. Worden; Nicholas C. Parazoo; Victoria Meyer; John T. Reager; Helen M. Worden; Zhe Jiang; Gregory R. Quetin; T. Luke Smallman; Jean-François Exbrayat; Yi Yin; Sassan S. Saatchi; Mathew Williams; David S. Schimel

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Open Access: N/A

Abstract: . Inter-annual variations in the tropical land carbon (C) balance are a dominant component of the global atmospheric CO₂ growth rate. Currently, the lack of quantitative knowledge on processes controlling net tropical ecosystem C balance on inter-annual timescales inhibits accurate understanding and projections of land-atmosphere C exchanges. In particular, uncertainty on the relative contribution of ecosystem C fluxes attributable to concurrent forcing anomalies (concurrent effects) and those attributable to the continuing influence of past phenomena (lagged effects) stifles efforts to explicitly understand the integrated sensitivity of a tropical ecosystem to climatic variability. Here we present a conceptual framework - applicable in principle to any land biosphere model - to explicitly quantify net biospheric exchange (NBE) as the sum of anomaly-induced concurrent changes and climatology-induced lagged changes to terrestrial ecosystem C states ($NBE = NBECON + NBELAG$). We apply this framework to an observation-constrained analysis of the 2001-2015 tropical C balance: we use a data-model integration approach (CARbon DATA-MODEL framework - CARDAMOM) to merge satellite-retrieved land-surface C observations (leaf area, biomass, solar-induced fluorescence), soil C inventory data and satellite-based atmospheric inversion estimates of CO₂ and CO fluxes to produce a data-constrained analysis of the 2001-2015 tropical C cycle. We find that the inter-annual variability of both concurrent and lagged effects substantially contributes to the 2001-2015 NBE inter-annual variability throughout 2001-2015 across the tropics ($NBECON$ IAV = 80 % of total NBE IAV, $r = 0.76$; $NBELAG$ IAV = 64 % of NBE IAV, $r = 0.61$), and the prominence of $NBELAG$ IAV persists across both wet and dry tropical ecosystems. The magnitude of lagged effect variations on NBE across the tropics is largely attributable to lagged effects on net primary productivity (NPP; $NPPLAG$ IAV 113 % of $NBELAG$ IAV, $r = -0.93$, p value < 0.05), which emerge due to the dependence of NPP on inter-annual variations in foliar C and plant-available H₂O states. We conclude that concurrent and lagged effects need to be explicitly and jointly resolved to retrieve an accurate understanding of the processes

regulating the present-day and future trajectory of the terrestrial land C sink.

68. Exploring constraints on a wetland methane emission ensemble (WetCHARTs) using GOSAT observations

Authors: Robert J. Parker; Chris Wilson; A. Anthony Bloom; Edward Comyn-Platt; Garry Hayman; Joe McNorton; Hartmut Boesch; Martyn P. Chipperfield

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Abstract: . Wetland emissions contribute the largest uncertainties to the current global atmospheric CH₄ budget, and how these emissions will change under future climate scenarios is also still poorly understood. Bloom et al. (2017b) developed WetCHARTs, a simple, data-driven, ensemble-based model that produces estimates of CH₄ wetland emissions constrained by observations of precipitation and temperature. This study performs the first detailed global and regional evaluation of the WetCHARTs CH₄ emission model ensemble against 9 years of high-quality, validated atmospheric CH₄ observations from GOSAT (the Greenhouse Gases Observing Satellite). A 3-D chemical transport model is used to estimate atmospheric CH₄ mixing ratios based on the WetCHARTs emissions and other sources. Across all years and all ensemble members, the observed global seasonal-cycle amplitude is typically underestimated by WetCHARTs by -7.4 ppb, but the correlation coefficient of 0.83 shows that the seasonality is well-produced at a global scale. The Southern Hemisphere has less of a bias (-1.9 ppb) than the Northern Hemisphere (-9.3 ppb), and our findings show that it is typically the North Tropics where this bias is the worst (-11.9 ppb). We find that WetCHARTs generally performs well in reproducing the observed wetland CH₄ seasonal cycle for the majority of wetland regions although, for some regions, regardless of the ensemble configuration, WetCHARTs does not reproduce the observed seasonal cycle well. In order to investigate this, we performed detailed analysis of some of the more challenging exemplar regions (Paraná River, Congo, Sudd and Yucatán). Our results show that certain ensemble members are more suited to specific regions, due to either deficiencies in the underlying data driving the model or complexities in representing the processes involved. In particular, incorrect definition of the wetland extent is found to be the most common reason for the discrepancy between the modelled and observed CH₄ concentrations. The remaining driving data (i.e. heterotrophic respiration and temperature) are shown to also contribute to the mismatch with observations, with the details differing on a region-by-region basis but generally showing that some degree of temperature dependency is better than none. We conclude that the data-driven approach used by WetCHARTs is

well-suited to producing a benchmark ensemble dataset against which to evaluate more complex process-based land surface models that explicitly model the hydrological behaviour of these complex wetland regions.

69. A spatial emergent constraint on the sensitivity of soil carbon turnover to global warming

Authors: Rebecca M. Varney; Sarah E. Chadburn; Pierre Friedlingstein; Eleanor J. Burke; Charles D. Koven; Gustaf Hugelius; Peter M. Cox

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Abstract: Carbon cycle feedbacks represent large uncertainties in climate change projections, and the response of soil carbon to climate change contributes the greatest uncertainty to this. Future changes in soil carbon depend on changes in litter and root inputs from plants and especially on reductions in the turnover time of soil carbon (τ_{soil}) with warming. An approximation to the latter term for the top one metre of soil ($\Delta C_{\text{soil},\tau_{\text{soil}}}$) can be diagnosed from projections made with the CMIP6 and CMIP5 Earth System Models (ESMs), and is found to span a large range even at 2 degreesC of global warming (-196 +/- 117 PgC). Here, we present a constraint on $\Delta C_{\text{soil},\tau_{\text{soil}}}$, which makes use of current heterotrophic respiration and the spatial variability of τ_{soil} inferred from observations. This spatial emergent constraint allows us to halve the uncertainty in $\Delta C_{\text{soil},\tau_{\text{soil}}}$ at 2 degreesC to -232 +/- 52 PgC.

70. Definitions and methods to estimate regional land carbon fluxes for the second phase of the REgional Carbon Cycle Assessment and Processes Project (RECCAP-2)

Authors: Philippe Ciais; Ana Bastos; Frédéric Chevallier; Ronny Lauerwald; Ben Poulter; Pep Canadell; Gustaf Hugelius; Robert B. Jackson; Atul Jain; Matthew Jones; Masayuki Kondo; Ingrid Luijkx; Prabir K. Patra; Wouter Peters; Julia Pongratz; A. M. Roxanna Petrescu; Shilong Piao; Chunjing Qiu; Celso Von Randow; Pierre Regnier; Marielle Saunois; Robert Scholes; Anatoli Shvidenko; Hanqin Tian; Hui Yang; Xuhui Wang; Bo Zheng

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Abstract: . Regional land carbon budgets provide insights on the spatial distribution of the land uptake of atmospheric carbon dioxide, and can be used to evaluate carbon cycle models and to define baselines for land-based additional mitigation efforts. The scientific community has been involved in providing observation-based estimates of regional carbon budgets either by downscaling atmospheric CO₂ observations into surface fluxes with atmospheric inversions, by using inventories of carbon stock changes in terrestrial ecosystems, by upscaling local field observations such as flux towers with gridded climate and remote sensing fields or by integrating data-driven or process-oriented terrestrial carbon cycle models. The first coordinated attempt to collect regional carbon budgets for nine regions covering the entire globe in the RECCAP-1 project has delivered estimates for the decade 2000-2009, but these budgets were not comparable between regions, due to different definitions and component fluxes reported or omitted. The recent recognition of lateral fluxes of carbon by human activities and rivers, that connect CO₂ uptake in one area with its release in another also requires better definition and protocols to reach harmonized regional budgets that can be summed up to the globe and compared with the atmospheric CO₂ growth rate and inversion results. In this study, for the international initiative RECCAP-2 coordinated by the Global Carbon Project, which aims as an update of regional carbon budgets over the last two decades based on observations, for 10 regions covering the globe, with a better harmonization than the precursor project, we provide recommendations for using atmospheric inversions results to match bottom-up carbon accounting and models, and we define the different component fluxes of the net land atmosphere carbon exchange that should be reported by each research group in charge of each region. Special attention is given to lateral fluxes, inland water

fluxes and land use fluxes.

71. A model-data fusion approach to analyse carbon dynamics in managed grasslands

Authors: Vasileios Myrgiotis; Emanuel Blei; Rob Clement; Stephanie K. Jones; Ben Keane; Mark A. Lee; Peter E. Levy; Robert M. Rees; Ute M. Skiba; Thomas Luke Smallman; Sylvia Toet; Mathew Williams

Journal: Agricultural Systems

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Abstract: N/A

72. Carbon Stocks and Fluxes in Kenyan Forests and Wooded Grasslands Derived from Earth Observation and Model-Data Fusion

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Keywords: Deforestation, multispectral, Random forests, Biomass, Carbon balance (and 5 more)

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Open Access: N/A

Abstract: The characterization of carbon stocks and dynamics at the national level is critical for countries engaging in climate change mitigation and adaptation strategies. However, several tropical countries, including Kenya, lack the essential information typically provided by a complete national forest inventory. Here we present the most detailed and rigorous national-scale assessment of aboveground woody biomass carbon stocks and dynamics for Kenya to date. A non-parametric random forest algorithm was trained to retrieve aboveground woody biomass carbon (AGBC) for the year 2014 $\pm$ 1 and forest disturbances for the 2014-2017 period using in situ forest inventory plot data and satellite Earth Observation (EO) data. The ecosystem carbon cycling of Kenya's forests and wooded grassland were assessed using a model-data fusion framework, CARDAMOM, constrained by the woody biomass datasets from this study as well as time series information on leaf area, fire events and soil organic carbon. Our EO-derived AGBC stocks were estimated as 140 Mt C for forests and 199 Mt C for wooded grasslands. The total AGBC loss during the study period was estimated as 1.89 Mt C with a dispersion below 1%. The CARDAMOM analysis estimated woody productivity to be three times larger in forests (mean = 1.9 t C ha⁻¹ yr⁻¹) than wooded grasslands (0.6 t C ha⁻¹ yr⁻¹), and the mean residence time of woody C in forests (16 years) to be greater than in wooded grasslands (10 years). This study stresses the importance of carbon sequestration by forests in the international climate

mitigation efforts under the Paris Agreement, but emphasizes the need to include non-forest ecosystems such as wooded grasslands in international greenhouse gas accounting frameworks.

73. The influence of two land-surface hydrology schemes on the terrestrial carbon cycle of Africa: A regional climate model study

Authors: Samy A. Anwar (Egyptian Meteorological Authority); Ismaila Diallo (Department of Geography University of California--Los Angeles Los Angeles California USA)

Journal: International Journal of Climatology

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Abstract: Two 16-year simulations were conducted to study the influence of two runoff schemes [the default one is TOPMODEL (hereafter referred as TOP) and the other one is the variable infiltration capacity (VIC)] with vegetation is being interactive [by enabling the carbon-nitrogen (CN) module] on the terrestrial carbon cycle fluxes of Africa using the regional climate model--RegCM4. The first simulation was labelled as CN-TOP, while the second simulation was hereafter referred to as CN-VIC. The results were evaluated and compared against various observation-based satellites and reanalysis products. Results showed that the CN-TOP overestimates leaf area index (LAI) and slightly underestimates the terrestrial carbon fluxes in comparison with the reference product; meanwhile, the CN-VIC severely underestimates both LAI and terrestrial fluxes particularly over the Congo basin. In addition, the CN-VIC shows a superior performance for simulating the net ecosystem exchange (NEE) than when it is simulated by the CN-TOP. In summary, the coupled RegCM4-CLM45-CN-VIC model can be recommended for future studies of NEE over Africa. To obtain robust results for simulating the terrestrial carbon fluxes especially the NEE; it is highly recommended to calibrate the four parameters of the VIC surface data set over Africa particularly over the Congo basin against in-situ observations.

74. Empirical estimates of regional carbon budgets imply reduced global soil heterotrophic respiration

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Abstract: Resolving regional carbon budgets is critical for informing land-based mitigation policy. For nine regions covering nearly the whole globe, we collected inventory estimates of carbon-stock changes complemented by satellite estimates of biomass changes where inventory data are missing. The net land-atmospheric carbon exchange (NEE) was calculated by taking the sum of the carbon-stock change and lateral carbon fluxes from crop and wood trade, and riverine-carbon export to the ocean. Summing up NEE from all regions, we obtained a global 'bottom-up' NEE for net land anthropogenic CO₂ uptake of -2.2 ± 0.6 PgC yr⁻¹ consistent with the independent top-down NEE from the global atmospheric carbon budget during 2000-2009. This estimate is so far the most comprehensive global bottom-up carbon budget accounting, which set up an important milestone for global carbon-cycle studies. By decomposing NEE into component fluxes, we found that global soil heterotrophic respiration amounts to a source of CO₂ of 39 PgC yr⁻¹ with an interquartile of 33-46 PgC yr⁻¹--a much smaller portion of net primary productivity than previously reported.

75. Carbon Flux Variability From a Relatively Simple Ecosystem Model With Assimilated Data Is Consistent With Terrestrial Biosphere Model Estimates

Authors: Gregory R. Quetin (Department of Earth System Science Stanford University Stanford CA USA); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology, University of California Los Angeles); Alexandra G. Konings (Department of Earth System Science Stanford University Stanford CA USA)

Journal: Journal of Advances in Modeling Earth Systems

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Keywords: data assimilation, carbon cycle, carbon modeling

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Open Access: N/A

Abstract: Modeling the net carbon balance is challenging due to the knowledge gaps in the variability and processes controlling gross carbon fluxes. Terrestrial carbon cycle modeling is susceptible to several sources of bias, including meteorological uncertainty, model structural uncertainty, and model parametric uncertainty. To determine the impact of these uncertainties, we compare three model-derived representations of the global terrestrial carbon balance across 1997-2009: (1) observation-constrained model-data fusion (CARBOn Data Model FraMEwork, CARDAMOM), (2) the reanalysis-driven Trends in Net Land-Atmosphere Carbon Exchange (TRENDY) land biosphere model ensemble, and (3) the Coupled Model Intercomparison Project 5 (CMIP5) Earth System Model ensemble. We consider the spread in carbon cycle simulations attributable primarily to parametric uncertainty (CARDAMOM), structural uncertainty (TRENDY), and combined structural and simulated meteorological uncertainty (CMIP5). We find that the spread across the CARDAMOM ensemble long-term mean--produced by parameter uncertainty--is larger than the spread of TRENDY and CMIP5 for net biosphere exchange (NBE) but similar for gross primary productivity (GPP). The carbon flux dynamics of CARDAMOM compares to models in TRENDY as well as models in TRENDY compare to each other in many regions for NBE seasonal (nine of 12), NBE interannual (11 of 12), and GPP seasonal variability (7 of 12), although not for GPP interannual variability (2 of 12). The simple model structure of CARDAMOM and systematic assimilation of observations is sufficient to produce carbon dynamics within the range of more complex models. These results are encouraging for the use of model-data fusion products with empirically estimated uncertainty for global carbon cycle studies.

76. Carbon dynamics in the Brazilian Cerrado: stocks and fluxes estimated by a model data fusion framework (CARDAMOM)

Authors: Eráclito Sousa-Neto; Luke Smallman; Jean Ometto; Mathew Williams

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Abstract:

Savannas are a major component of the world's vegetation and cover a land surface of about 15 Mkm², accounting for about 30% of the terrestrial primary production. In the South America, the Brazilian Savanna (Cerrado) is the second largest biome (2 Mkm²), after the Amazon biome, and a hotspot of biodiversity. The Cerrado region is heterogeneous, with savanna vegetation ranging from open grassland, through a gradient of increasing tree density to nearly closed-canopy woodland. The cerrado vegetation is markedly seasonal in phenology and is often burned, either naturally or as part of a management cycle. Due its large occupation, Cerrado have the potential to influence the regional and possibly the global energy, water and carbon (C) balances. The allocation of the net primary productivity (NPP) of an ecosystem between canopy, woody tissue and fine roots is an important descriptor of the functioning of an ecosystem, and an important feature to correctly represent in terrestrial ecosystem models for carbon rates estimation, as well as their residence time, variation with climate and disturbance, and in order to make better forecasts. Such estimation in Cerrado regions remains still difficult given the lack of important soil and vegetation data. Previous studies have showed that the fluxes of water and C are closely related to each other, and to the diurnal cycle of solar radiation. However, there is no study clearly assessing the allocation of C through the different types of vegetation, either in the different types of physiognomies. To help estimating the C flows across the different C pools and types of vegetation, we are using Carbon Data Model Framework (CARDAMOM) which is a computer programme that retrieves terrestrial carbon (C) cycle variables by combining C cycle observations with a mass balance model. CARDAMOM produces global dynamic estimates of plant and soil C pools, their exchanges with each other and with the atmosphere, and C cycling variables for processes driving change. It also produces a C cycle analysis consistent with C measurements and climate, and it is suited for using with global-scale satellite observations such as aboveground biomass (AGB) or leaf area index (LAI). For that, we count on field data available (AGB, BGB) and satellite data (LAI, AGB, soil C), which will help to present robust analyses of C cycling across gradients of biomass in the Brazilian Cerrado.

77. Optimal model complexity for terrestrial carbon cycle prediction using data assimilation

Authors: Caroline A. Famiglietti; T. Luke Smallman; Sophie Flack-Prain; Rong Ge; Victoria Meyer; Nicholas C. Parazoo; Gregory R. Quetin; Andrew Revill; Stephanie G. Stettz; Yan Yang; Yuan Zhao; Penghui Zhu; A. Anthony Bloom; Mathew Williams; Alexandra G. Konings

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Subjects: N/A

Keywords: Structural complexity|Carbon flux|Ecosystem model|Equifinality|Predictability|Biosphere model|Terrestrial ecosystem

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Abstract: <p>The future role of the terrestrial biosphere in the global carbon cycle is highly uncertain. Modeling and predicting the terrestrial net carbon balance is difficult due to the numerous processes driving variability of gross fluxes. Many approaches to reducing this model uncertainty have focused on model structure, namely by adding additional processes (e.g., nutrient dynamics or vegetation demography) and thus increasing complexity. While these developments seek to achieve greater structural realism by mirroring the complexity of the natural world, they often rely, by necessity, on poorly-determined or over-generalized parameters. Furthermore, increased structural complexity may increase the risk that parameters with compensating errors are found during model development, thereby reducing model accuracy in prediction. It is not clear whether or to what extent carbon cycle predictability scales with structural complexity, or whether an intermediate, optimum level of complexity exists that may balance the costs of a low (more biased) or high (more variant) complexity model. Here, we explore and define the relationship between carbon cycle model complexity and prediction accuracy. To do so, we leverage the CARbon Data MOdel fraMework (CARDAMOM), a Bayesian data assimilation system that retrieves terrestrial carbon cycle variables (including pools, fluxes, and static parameters) by combining multiple observations with a relatively simple ecosystem carbon balance model. CARDAMOM includes several ecological and dynamical constraints that can prevent ecologically unrealistic parameter combinations and reduce compensating errors between parameters (also known as equifinality). Furthermore, it is a flexible framework to which process representations, parameters, and constraints can easily be added and removed. We used CARDAMOM to develop a suite of model versions spanning a broad range of structural complexity, including the number of carbon pools and the allocation of carbon to the canopy. We assessed a model's complexity based on its inherent dimensionality, determined via a principal component analysis that reduces the parameter space to its principal components. We tested and compared the training and forecast

accuracies of net ecosystem exchange predictions using 14 increasingly complex versions of CARDAMOM, each with 48 different experimental designs (*i.e.*, combinations of data constraints and error assumptions) at 5 globally-distributed eddy covariance sites representing a range of biomes and vegetation types across a total of 70 site-years. We also compared the model performance values against a range of machine learning approaches, which are assumed to represent the limit of infinite model complexity due to their large number of underlying parameters. In this presentation, we use this population to demonstrate and explain patterns in the mapping of model complexity and other assimilation choices to prediction accuracy, offering theoretical and empirical insights into the optimal structure of a carbon cycle model.

78. A spatial emergent constraint on the sensitivity of soil carbon turnover time to global warming

Authors: Rebecca Varney; Peter Cox; Sarah Chadburn; Pierre Friedlingstein; Eleanor Burke; Charles Koven; Gustaf Hugelius

Journal: Unknown Journal

Publisher: Copernicus GmbH

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Abstract:

Carbon cycle feedbacks represent large uncertainties on climate change projections, and the response of soil carbon to climate change contributes the greatest uncertainty to this. Future changes in soil carbon depend on changes in litter and root inputs from plants, and especially on reductions in the turnover time of soil carbon (τ) with warming. The latter represents the change in soil carbon due to the response of soil turnover time (τ), and can be diagnosed from projections made with Earth System Models (ESMs). It is found to span a large range even at the Paris Agreement Target of 2°C global warming. We use the spatial variability of τ inferred from observations to obtain a constraint on τ . This spatial emergent constraint allows us to greatly reduce the uncertainty in τ at 2°C global warming. We do likewise for other levels of global warming to derive a best estimate for the effective sensitivity of τ to global warming, and derive a q10 equivalent value for heterotrophic respiration.

79. Lagged effects dominate the inter-annual variability of the 2010-2015 tropical carbon balance

Authors: A. Anthony Bloom; Kevin W. Bowman; Junjie Liu; Alexandra G. Konings; John R. Worden; Nicholas C. Parazoo; Victoria Meyer; John T. Reager; Helen M. Worden; Zhe Jiang; Gregory R. Quetin; T. Luke Smallman; Jean-François Exbrayat; Yi Yin; Sassan S. Saatchi; Mathew Williams; David S. Schimel

Journal: Unknown Journal

Publisher: Copernicus GmbH

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References: N/A

Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: Biosphere model|Land Cover|Dominance (genetics)

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . Inter-annual variations in the tropical land carbon (C) balance are a dominant component of the global atmospheric CO₂ growth rate. Currently, the lack of quantitative knowledge on processes controlling net tropical ecosystems C balance on inter-annual timescales inhibits accurate understanding and projections of land-atmosphere C exchanges. In particular, uncertainty on the relative contribution of ecosystem C fluxes attributable to concurrent meteorological forcing anomalies (concurrent effects) and those attributable to the continuing influence of past phenomena (lagged effects) stifles efforts to explicitly understand the integrated sensitivity of tropical ecosystem to climatic variability. Here we present a conceptual framework - applicable in principle to any meteorology-forced land biosphere model - to explicitly quantify net biospheric exchange (NBE) as the sum of anomaly-induced concurrent changes and climatology-induced lagged changes to terrestrial ecosystem C states ($NBE = NBECON + NBELAG$). We apply this framework to an observation-constrained analysis of the 2010-2015 tropical C balance: we use a data-model integration approach (CARDAMOM) to merge satellite-retrieved land-surface C observations (leaf area, biomass, solar-induced fluorescence), soil C inventory data and satellite-based atmospheric inversion estimates of CO₂ and CO fluxes to produce a data-constrained analysis of the 2010-2015 tropical C cycle. We find that the inter-annual variability of lagged effects explain the majority of NBE inter-annual variability (IAV) throughout 2010-2015 across the tropics ($NBELAG$ IAV = 112 % of NBE IAV, $r = 0.87$) relative to concurrent effects ($NBECON$ IAV = 54 % of total NBE IAV, $r = 0.03$) and the dominance of $NBELAG$ IAV persists across both wet and dry tropical ecosystems. The magnitude of lagged effect variations on NBE across the tropics is largely attributable to lagged effects on net primary productivity (NPP; $NPPLAG$ IAV 88 % of $NBELAG$ IAV, $r = -0.99$, p-value

80. Description and validation of an intermediate complexity model for ecosystem photosynthesis and evapotranspiration: ACM-GPP-ETv1

Authors: Thomas Luke Smallman; Mathew Williams

Journal: Geoscientific Model Development

Publisher: Copernicus GmbH

DOI: 10.5194/gmd-12-2227-2019

URL: <https://doi.org/10.5194/gmd-12-2227-2019>

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Volume/Issue: Volume 12, Issue 6

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Cited by: 18 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council, UK Space Agency

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Open Access: N/A

Abstract: . Photosynthesis (gross primary production, GPP) and evapotranspiration (ET) are ecosystem processes with global significance for climate, the global carbon and hydrological cycles and a range of ecosystem services. The mechanisms governing these processes are complex but well understood. There is strong coupling between these processes, mediated directly by stomatal conductance and indirectly by root zone soil moisture content and its accessibility. This coupling must be effectively modelled for robust predictions of earth system responses to global change. Yet, it is highly demanding to model leaf and cellular processes, like stomatal conductance or electron transport, with response times of minutes, over decadal and global domains. Computational demand means models resolving this level of complexity cannot be easily evaluated for their parameter sensitivity nor calibrated using earth observation information through data assimilation approaches requiring large ensembles. To overcome these challenges, here we describe a coupled photosynthesis evapotranspiration model of intermediate complexity. The model reduces computational load and parameter numbers by operating at canopy scale and daily time step. Through the inclusion of simplified representation of key process interactions, it retains sensitivity to variation in climate, leaf traits, soil states and atmospheric CO₂. The new model is calibrated to match the biophysical responses of a complex terrestrial ecosystem model (TEM) of GPP and ET through a Bayesian model-data fusion framework. The calibrated ACM-GPP-ET generates unbiased estimates of TEM GPP and ET and captures 80 %-95 % of the sensitivity of carbon and water fluxes by the complex TEM. The ACM-GPP-ET model operates 3 orders faster than the complex TEM. Independent evaluation of ACM-GPP-ET at FLUXNET sites, using a single global parameterisation, shows good agreement, with typical R²~0.60 for both GPP and ET. This intermediate complexity modelling approach allows full Monte Carlo-based quantification of model parameter and structural uncertainties and global-scale sensitivity analyses for these processes and is fast enough for use within terrestrial ecosystem model-data fusion frameworks requiring large ensembles.

81. Global satellite-driven estimates of heterotrophic respiration

Authors: Alexandra G. Konings; A. Anthony Bloom; Junjie Liu; Nicholas C. Parazoo; David S. Schimel; Kevin W. Bowman

Journal: Biogeosciences

Publisher: Copernicus GmbH

DOI: 10.5194/bg-16-2269-2019

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Publication Date: 2019-06-04

Volume/Issue: Volume 16, Issue 11

References: 98

Cited by: 37 publications

Language: en

Subjects: N/A

Keywords: N/A

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License: CC BY 4.0

Open Access: N/A

Abstract: . While heterotrophic respiration (Rh) makes up about a quarter of gross global terrestrial carbon fluxes, it remains among the least-observed carbon fluxes, particularly outside the midlatitudes. In situ measurements collected in the Soil Respiration Database (SRDB) number only a few hundred worldwide. Similarly, only a single data-driven wall-to-wall estimate of annual average heterotrophic respiration exists, based on bottom-up upscaling of SRDB measurements using an assumed functional form to account for climate variability. In this study, we exploit recent advances in remote sensing of terrestrial carbon fluxes to estimate global variations in heterotrophic respiration in a top-down fashion at monthly temporal resolution and 4°x5° spatial resolution. We combine net ecosystem productivity estimates from atmospheric inversions of the NASA Carbon Monitoring System-Flux (CMS-Flux) with an optimally scaled gross primary productivity dataset based on satellite-observed solar-induced fluorescence variations to estimate total ecosystem respiration as a residual of the terrestrial carbon balance. The ecosystem respiration is then separated into autotrophic and heterotrophic components based on a spatially varying carbon use efficiency retrieved in a model-data fusion framework (the CARbon DAta MOdel fraMework, CARDAMOM). The resulting dataset is independent of any assumptions about how heterotrophic respiration responds to climate or substrate variations. It estimates an annual average global average heterotrophic respiration flux of 43.6±19.3 Pg C yr⁻¹. Sensitivity and uncertainty analyses showed that the top-down Rh are more sensitive to the choice of input gross primary productivity (GPP) and net ecosystem productivity (NEP) datasets than to the assumption of a static carbon use efficiency (CUE) value, with the possible exception of the wet tropics. These top-down estimates are compared to bottom-up estimates of annual heterotrophic respiration, using new uncertainty estimates that partially account for sampling and model errors. Top-down heterotrophic respiration estimates are higher than those from bottom-up upscaling everywhere except at high latitudes and are 30 % greater overall (43.6 Pg C yr⁻¹ vs. 33.4 Pg C yr⁻¹). The uncertainty ranges of both methods are comparable, except poleward of 45° N, where bottom-up uncertainties are greater. The ratio of top-down heterotrophic to total ecosystem respiration varies seasonally by as much as 0.6 depending on season and climate, illustrating the importance of

studying the drivers of autotrophic and heterotrophic respiration separately, and thus the importance of data-driven estimates of R_h such as those estimated here.

82. Evaluation of terrestrial pan-Arctic carbon cycling using a data-assimilation system

Authors: Efrén López-Blanco; Jean-François Exbrayat; Magnus Lund; Torben R. Christensen; Mikkel P. Tamstorf; Darren Slevin; Gustaf Hugelius; Anthony A. Bloom; Mathew Williams

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

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Volume/Issue: Volume 10, Issue 2

References: 115

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: CC BY 4.0

Open Access: N/A

Abstract: . There is a significant knowledge gap in the current state of the terrestrial carbon (C) budget. Recent studies have highlighted a poor understanding particularly of C pool transit times and of whether productivity or biomass dominate these biases. The Arctic, accounting for approximately 50 % of the global soil organic C stocks, has an important role in the global C cycle. Here, we use the CARbon Data Model (CARDAMOM) data-assimilation system to produce pan-Arctic terrestrial C cycle analyses for 2000-2015. This approach avoids using traditional plant functional type or steady-state assumptions. We integrate a range of data (soil organic C, leaf area index, biomass, and climate) to determine the most likely state of the high-latitude C cycle at a 1° x 1° resolution and also to provide general guidance about the controlling biases in transit times. On average, CARDAMOM estimates regional mean rates of photosynthesis of 565 g C m⁻² yr⁻¹ (90 % confidence interval between the 5th and 95th percentiles: 428, 741), autotrophic respiration of 270 g C m⁻² yr⁻¹ (182, 397) and heterotrophic respiration of 219 g C m⁻² yr⁻¹ (31, 1458), suggesting a pan-Arctic sink of -67 (-287, 1160) g C m⁻² yr⁻¹, weaker in tundra and stronger in taiga. However, our confidence intervals remain large (and so the region could be a source of C), reflecting uncertainty assigned to the regional data products. We show a clear spatial and temporal agreement between CARDAMOM analyses and different sources of assimilated and independent data at both pan-Arctic and local scales but also identify consistent biases between CARDAMOM and validation data. The assimilation process requires clearer error quantification for leaf area index (LAI) and biomass products to resolve these biases. Mapping of vegetation C stocks and change over time and soil C ages linked to soil C stocks is required for better analytical constraint. Comparing CARDAMOM analyses to global vegetation models (GVMs) for the same period, we conclude that transit times of vegetation C are inconsistently simulated in GVMs due to a combination of uncertainties from productivity and biomass calculations. Our findings highlight that GVMs need to focus on constraining both current vegetation C stocks and net primary production to improve a process-based understanding of C cycle dynamics in the Arctic.

83. Quantifying the UK's carbon dioxide flux: an atmospheric inverse modelling approach using a regional measurement network

Authors: Emily D. White; Matthew Rigby; Mark F. Lunt; T. Luke Smallman; Edward Comyn-Platt; Alistair J. Manning; Anita L. Ganesan; Simon O'Doherty; Ann R. Stavert; Kieran Stanley; Mathew Williams; Peter Levy; Michel Ramonet; Grant L. Forster; Andrew C. Manning; Paul I. Palmer

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-19-4345-2019

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Publication Date: 2019-04-04

Volume/Issue: Volume 19, Issue 7

References: 69

Cited by: 20 publications

Language: en

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Keywords: N/A

Funding: Department for Business, Energy and Industrial Strategy, UK Government

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Open Access: N/A

Abstract: . We present a method to derive atmospheric-observation-based estimates of carbon dioxide (CO₂) fluxes at the national scale, demonstrated using data from a network of surface tall-tower sites across the UK and Ireland over the period 2013–2014. The inversion is carried out using simulations from a Lagrangian chemical transport model and an innovative hierarchical Bayesian Markov chain Monte Carlo (MCMC) framework, which addresses some of the traditional problems faced by inverse modelling studies, such as subjectivity in the specification of model and prior uncertainties. Biospheric fluxes related to gross primary productivity and terrestrial ecosystem respiration are solved separately in the inversion and then combined a posteriori to determine net ecosystem exchange of CO₂. Two different models, Data Assimilation Linked Ecosystem Carbon (DALEC) and Joint UK Land Environment Simulator (JULES), provide prior estimates for these fluxes. We carry out separate inversions to assess the impact of these different priors on the posterior flux estimates and evaluate the differences between the prior and posterior estimates in terms of missing model components. The Numerical Atmospheric Dispersion Modelling Environment (NAME) is used to relate fluxes to the measurements taken across the regional network. Posterior CO₂ estimates from the two inversions agree within estimated uncertainties, despite large differences in the prior fluxes from the different models. With our method, averaging results from 2013 and 2014, we find a total annual net biospheric flux for the UK of 8 ± 79 Tg CO₂ yr⁻¹ (DALEC prior) and 64 ± 85 Tg CO₂ yr⁻¹ (JULES prior), where negative values represent an uptake of CO₂. These biospheric CO₂ estimates show that annual UK biospheric sources and sinks are roughly in balance. These annual mean estimates consistently indicate a greater net release of CO₂ than the prior estimates, which show much more pronounced uptake in summer months.

84. Understanding the Land Carbon Cycle with Space Data: Current Status and Prospects

Authors: Jean-François Exbrayat; A. Anthony Bloom; Nuno Carvalhais; Rico Fischer; Andreas Huth; Natasha MacBean; Mathew Williams

Journal: Surveys in Geophysics

Publisher: Springer Science and Business Media LLC

DOI: 10.1007/s10712-019-09506-2

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Publication Date: 2019-02-06

Volume/Issue: Volume 40, Issue 4

References: 168

Cited by: 21 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: N/A

License: N/A

Open Access: N/A

Abstract: N/A

85. Inverse Determination of the Influence of Fire on Vegetation Carbon Turnover in the Pantropics

Authors: Jean-François Exbrayat (University of Edinburgh); T. Luke Smallman (University of Edinburgh); A. Anthony Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Lindsay B. Hutley (Charles Darwin University); Mathew Williams (University of Edinburgh)

Journal: Global Biogeochemical Cycles

Publisher: American Geophysical Union (AGU)

DOI: 10.1029/2018gb005925

URL: <https://doi.org/10.1029/2018gb005925>

Publication Date: 2018-12-03

Volume/Issue: Volume 32, Issue 12

References: 72

Cited by: 20 publications

Language: en

Subjects: N/A

Keywords: model-data fusion, tropics, sensitivity, terrestrial carbon cycle, inverse modeling (and 1 more)

Funding: Jet Propulsion Laboratory, Natural Environment Research Council, Scottish Alliance for Geoscience, Environment and Society (and 3 more)

License: N/A

Open Access: N/A

Abstract: Fire is a major component of the terrestrial carbon cycle that has been implemented in most current global terrestrial ecosystem models. Here we use terrestrial carbon cycle observations to characterize the importance of fire regime gradients in the spatial distribution of ecosystem functional properties such as carbon allocation, fluxes, and turnover times in the tropics. A Bayesian model-data fusion approach is applied to an ecosystem carbon model to derive the posterior distribution of corresponding parameters for the tropics from 2000 to 2015. We perform the model-data fusion procedure twice, that is, with and without imposing fire. Gradient of differences in model parameters and ecosystem properties in response to fire emerge between these experiments. For example, mean annual burned fraction correlates with an increase in carbon use efficiency and reductions in carbon turnover times. Further, our analyses reveal an increased allocation to more fire-resistant tissues in the most frequently burned regions. As fire modules are increasingly implemented in global terrestrial ecosystem models, we recommend that model development includes a representation of the impact of fire on ecosystem properties as they may lead to large differences under climate change projections.

86. GOLUM-CNP v1.0: a data-driven modeling of carbon, nitrogen and phosphorus cycles in major terrestrial biomes

Authors: Yilong Wang; Philippe Ciais; Daniel Goll; Yuanyuan Huang; Yiqi Luo; Ying-Ping Wang; A. Anthony Bloom; Grégoire Broquet; Jens Hartmann; Shushi Peng; Josep Penuelas; Shilong Piao; Jordi Sardans; Benjamin D. Stocker; Rong Wang; Sönke Zaehle; Sophie Zechmeister-Boltenstern

Journal: Geoscientific Model Development

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DOI: 10.5194/gmd-11-3903-2018

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References: 112

Cited by: 33 publications

Language: en

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Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Global terrestrial nitrogen (N) and phosphorus (P) cycles are coupled to the global carbon (C) cycle for net primary production (NPP), plant C allocation, and decomposition of soil organic matter, but N and P have distinct pathways of inputs and losses. Current C-nutrient models exhibit large uncertainties in their estimates of pool sizes, fluxes, and turnover rates of nutrients, due to a lack of consistent global data for evaluating the models. In this study, we present a new model-data fusion framework called the Global Observation-based Land-ecosystems Utilization Model of Carbon, Nitrogen and Phosphorus (GOLUM-CNP) that combines the CARbon DATA Model framework (CARDAMOM) data-constrained C-cycle analysis with spatially explicit data-driven estimates of N and P inputs and losses and with observed stoichiometric ratios. We calculated the steady-state N- and P-pool sizes and fluxes globally for large biomes. Our study showed that new N inputs from biological fixation and deposition supplied >20 % of total plant uptake in most forest ecosystems but accounted for smaller fractions in boreal forests and grasslands. New P inputs from atmospheric deposition and rock weathering supplied a much smaller fraction of total plant uptake than new N inputs, indicating the importance of internal Precycling within ecosystems to support plant growth. Nutrient-use efficiency, defined as the ratio of gross primary production (GPP) to plant nutrient uptake, were diagnosed from our model results and compared between biomes. Tropical forests had the lowest N-use efficiency and the highest P-use efficiency of the forest biomes. An analysis of sensitivity and uncertainty indicated that the NPP-allocation fractions to leaves, roots, and wood contributed the most to the uncertainties in the estimates of nutrient-use efficiencies. Correcting for biases in NPP-allocation fractions produced more plausible gradients of N- and P-use efficiencies from tropical to boreal ecosystems and highlighted the critical role of accurate measurements of C allocation for understanding the N and P cycles.

87. Evaluation of terrestrial pan-Arctic carbon cycling using a data-assimilation system

Authors: Efrén López-Blanco; Jean-François Exbrayat; Magnus Lund; Torben R. Christensen; Mikkel P. Tamstorf; Darren Slevin; Gustaf Hugelius; Anthony A. Bloom; Mathew Williams

Journal: Unknown Journal

Publisher: Copernicus GmbH

DOI: 10.5194/esd-2018-19

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Publication Date: 2018-05-22

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Cited by: 1 publications

Language: N/A

Subjects: N/A

Keywords: Arctic vegetation|Terrestrial ecosystem

Funding: N/A

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Open Access: N/A

Abstract: . There is a significant knowledge gap in the current state of the terrestrial carbon (C) budget. The Arctic accounts for approximately 50 % of the global soil organic C stock, emphasizing the important role of Arctic regions in the global C cycle. Recent studies have pointed to the poor understanding of C pools turnover, although remain unclear as to whether productivity or biomass dominate the biases. Here, we use an improved version of the CARDAMOM data-assimilation system, to produce pan-Arctic terrestrial C-related variables without using traditional plant functional type or steady-state assumptions. Our approach integrates a range of data (soil organic C, leaf area index, biomass, and climate) to determine the most likely state of the high latitude C cycle at a 1 degrees x 1 degrees resolution for the first 15 years of the 21st century, but also to provide general guidance about the controlling biases in the turnover dynamics. As average, CARDAMOM estimates 513 (456, 579), 245 (208, 290) and 204 (109, 427) g C m⁻² yr⁻¹ (90 % confidence interval) from photosynthesis, autotrophic and heterotrophic respiration respectively, suggesting that the pan-Arctic region acted as a likely sink -55 (-152, 157) g C m⁻² yr⁻¹, weaker in tundra and stronger in taiga, but our confidence intervals remain large (and so the region could be a source of C). In general, we find a good agreement between CARDAMOM and different sources of assimilated and independent data at both pan-Arctic and local scale. Using CARDAMOM as a benchmarking tool for global vegetation models (GVM), we also conclude that turnover time of vegetation C is weakly simulated in vegetation models and is a major component of error in their forecasts. Our findings highlight that GVM modellers need to focus on the vegetation C stocks dynamics, but also their respiratory losses, to improve our process-based understanding of internal C cycle dynamics in the Arctic.

88. High-resolution inversion of methane emissions in the Southeast US using SEAC4RS aircraft observations of atmospheric methane: anthropogenic and wetland sources

Authors: Jian-Xiong Sheng; Daniel J. Jacob; Alexander J. Turner; Joannes D. Maasakkers; Melissa P. Sulprizio; A. Anthony Bloom; Arlyn E. Andrews; Debra Wunch

Journal: Atmospheric Chemistry and Physics

Publisher: Copernicus GmbH

DOI: 10.5194/acp-18-6483-2018

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References: 41

Cited by: 34 publications

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Funding: N/A

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Open Access: N/A

Abstract: . We use observations of boundary layer methane from the SEAC4RS aircraft campaign over the Southeast US in August-September 2013 to estimate methane emissions in that region through an inverse analysis with up to $0.25^{\circ} \times 0.3125^{\circ}$ (25×25 km²) resolution and with full error characterization. The Southeast US is a major source region for methane including large contributions from oil and gas production and wetlands. Our inversion uses state-of-the-art emission inventories as prior estimates, including a gridded version of the anthropogenic EPA Greenhouse Gas Inventory and the mean of the WetCHARTs ensemble for wetlands. Inversion results are independently verified by comparison with surface (NOAA/ESRL) and column (TCCON) methane observations. Our posterior estimates for the Southeast US are 12.8 ± 0.9 Tg a⁻¹ for anthropogenic sources (no significant change from the gridded EPA inventory) and 9.4 ± 0.8 Tg a⁻¹ for wetlands (27 % decrease from the mean in the WetCHARTs ensemble). The largest source of error in the WetCHARTs wetlands ensemble is the land cover map specification of wetland areal extent. Our results support the accuracy of the EPA anthropogenic inventory on a regional scale but there are significant local discrepancies for oil and gas production fields, suggesting that emission factors are more variable than assumed in the EPA inventory.

89. Dominant regions and drivers of the variability of the global land carbon sink across timescales

Authors: Xuanze Zhang (Peking University); Ying-Ping Wang (Chinese Academy of Sciences, CSIRO Oceans and Atmosphere Aspendale VIC Australia); Shushi Peng (Peking University); Peter J. Rayner (University of Melbourne); Philippe Ciais (Laboratoire des Sciences du Climat et de l'Environnement LSCE/IPSL CEA-CNRS-UVSQ Université Paris-Saclay Gif-sur-Yvette France); Jeremy D. Silver (University of Melbourne); Shilong Piao (Chinese Academy of Sciences, Peking University); Zaichun Zhu (Peking University); Xingjie Lu (University of Oklahoma); Xiaogu Zheng (Chinese Academy of Sciences)

Journal: Global Change Biology

Publisher: Wiley

DOI: 10.1111/gcb.14275

URL: <https://doi.org/10.1111/gcb.14275>

Publication Date: 2018-04-17

Volume/Issue: Volume 24, Issue 9

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Cited by: 34 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: National Natural Science Foundation of China, China Postdoctoral Science Foundation, Commonwealth Scientific and Industrial Research Organisation (and 1 more)

License: N/A

Open Access: N/A

Abstract: Net biome productivity (NBP) dominates the observed large variation of atmospheric CO₂ annual increase over the last five decades. However, the dominant regions controlling inter-annual to multi-decadal variability of global NBP are still controversial (semi-arid regions vs. temperate or tropical forests). By developing a theory for partitioning the variance of NBP into the contributions of net primary production (NPP) and heterotrophic respiration (Rh) at different timescales, and using both observation-based atmospheric CO₂ inversion product and the outputs of 10 process-based terrestrial ecosystem models forced by 110-year observational climate, we tried to reconcile the controversy by showing that semi-arid lands dominate the variability of global NBP at inter-annual (<10 years) and tropical forests dominate at multi-decadal scales (>30 years). Results further indicate that global NBP variability is dominated by the NPP component at inter-annual timescales, and is progressively controlled by Rh with increasing timescale. Multi-decadal NBP variations of tropical rainforests are modulated by the Pacific Decadal Oscillation (PDO) through its significant influences on both temperature and precipitation. This study calls for long-term observations for the decadal or longer fluctuations in carbon fluxes to gain insights on the future evolution of global NBP, particularly in the tropical forests that dominate the decadal variability of land carbon uptake and are more effective for climate mitigation.

90. GOLUM-CNP v1.0: a data-driven modeling of carbon, nitrogen and phosphorus cycles in major terrestrial biomes

Authors: Yilong Wang; Philippe Ciais; Daniel Goll; Yuanyuan Huang; Yiqi Luo; Ying-Ping Wang; A. Anthony Bloom; Grégoire Broquet; Jens Hartmann; Shushi Peng; Josep Penuelas; Shilong Piao; Jordi Sardans; Benjamin D. Stocker; Rong Wang; Sönke Zaehle; Sophie Zechmeister-Boltenstern

Journal: Unknown Journal

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DOI: 10.5194/gmd-2018-23

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Cited by: N/A publications

Language: N/A

Subjects: N/A

Keywords: N/A

Funding: N/A

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Open Access: N/A

Abstract: . Global terrestrial nitrogen (N) and phosphorus (P) cycles are coupled to the global carbon (C) cycle for net primary production (NPP), plant C allocation and decomposition of soil organic matter, but N and P have distinct pathways of inputs and losses. Current C-nutrient models exhibit large uncertainties in their estimates of pool sizes, fluxes and turnover rates of nutrients, due to a lack of consistent global data for evaluating the models. In this study, we present a new model-data fusion framework called Global Observation-based Land-ecosystems Utilization Model of Carbon, Nitrogen and Phosphorus (GOLUM-CNP) that combines the CARbon DATA MOdel fraMework (CARDAMOM) data-constrained C-cycle analysis with spatially explicit data-driven estimates of N and P inputs and losses and with observed stoichiometric ratios. We calculated the steady-state N- and P-pool sizes and fluxes globally for large biomes. Our study showed that new N inputs from biological fixation and deposition supplied > 20 % of total plant uptake in most forest ecosystems but accounted for smaller fractions in boreal forests and grasslands. New P inputs from atmospheric deposition and rock weathering supplied a much smaller fraction of total plant uptake than new N inputs, indicating that the terrestrial C sink may ultimately be constrained by low P. Nutrient-use efficiency, defined as the ratio of gross primary production (GPP) to plant nutrient uptake, can be diagnosed from our model results and compared between biomes. Tropical forests had the lowest N-use efficiency and the highest P-use efficiency of the forest biomes. An analysis of sensitivity and uncertainty indicated that the NPP-allocation fractions to leaves, roots and wood contributed the most to the uncertainties in the estimates of nutrient-use efficiencies. Correcting for biases in NPP-allocation fractions produced more plausible gradients of N- and P-use efficiencies from tropical to boreal ecosystems and highlighted the critical role of accurate measurements of C allocation for understanding the N and P cycles.

91. Reliability ensemble averaging of 21st century projections of terrestrial net primary productivity reduces global and regional uncertainties

Authors: Jean-François Exbrayat; A. Anthony Bloom; Pete Falloon; Akihiko Ito; T. Luke Smallman; Mathew Williams

Journal: Earth System Dynamics

Publisher: Copernicus GmbH

DOI: 10.5194/esd-9-153-2018

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Publication Date: 2018-02-21

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References: 80

Cited by: 26 publications

Language: en

Subjects: N/A

Keywords: N/A

Funding: Natural Environment Research Council

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Open Access: N/A

Abstract: . Multi-model averaging techniques provide opportunities to extract additional information from large ensembles of simulations. In particular, present-day model skill can be used to evaluate their potential performance in future climate simulations. Multi-model averaging methods have been used extensively in climate and hydrological sciences, but they have not been used to constrain projected plant productivity responses to climate change, which is a major uncertainty in Earth system modelling. Here, we use three global observationally orientated estimates of current net primary productivity (NPP) to perform a reliability ensemble averaging (REA) method using 30 global simulations of the 21st century change in NPP based on the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP) "business as usual" emissions scenario. We find that the three REA methods support an increase in global NPP by the end of the 21st century (2095-2099) compared to 2001-2005, which is 2-3 % stronger than the ensemble ISIMIP mean value of 24.2 Pg C y⁻¹. Using REA also leads to a 45-68 % reduction in the global uncertainty of 21st century NPP projection, which strengthens confidence in the resilience of the CO₂ fertilization effect to climate change. This reduction in uncertainty is especially clear for boreal ecosystems although it may be an artefact due to the lack of representation of nutrient limitations on NPP in most models. Conversely, the large uncertainty that remains on the sign of the response of NPP in semi-arid regions points to the need for better observations and model development in these regions.

92. Global evaluation of gross primary productivity in the JULES land surface model v3.4.1

Authors: Darren Slevin; Simon F. B. Tett; Jean-François Exbrayat; A. Anthony Bloom; Mathew Williams

Journal: Geoscientific Model Development

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Abstract: . This study evaluates the ability of the JULES land surface model (LSM) to simulate gross primary productivity (GPP) on regional and global scales for 2001-2010. Model simulations, performed at various spatial resolutions and driven with a variety of meteorological datasets (WFDEI-GPCC, WFDEI-CRU and PRINCETON), were compared to the MODIS GPP product, spatially gridded estimates of upscaled GPP from the FLUXNET network (FLUXNET-MTE) and the CARDAMOM terrestrial carbon cycle analysis. Firstly, when JULES was driven with the WFDEI-GPCC dataset (at 0.5 degrees x 0.5 degrees spatial resolution), the annual average global GPP simulated by JULES for 2001-2010 was higher than the observation-based estimates (MODIS and FLUXNET-MTE), by 25 and 8 %, respectively, and CARDAMOM estimates by 23 %. JULES was able to simulate the standard deviation of monthly GPP fluxes compared to CARDAMOM and the observation-based estimates on global scales. Secondly, GPP simulated by JULES for various biomes (forests, grasslands and shrubs) on global and regional scales were compared. Differences among JULES, MODIS, FLUXNET-MTE and CARDAMOM on global scales were due to differences in simulated GPP in the tropics. Thirdly, it was shown that spatial resolution (0.5 degrees x 0.5 degrees, 1 degrees x 1 degrees and 2 degrees x 2 degrees) had little impact on simulated GPP on these large scales, with global GPP ranging from 140 to 142 PgC year⁻¹. Finally, the sensitivity of JULES to meteorological driving data, a major source of model uncertainty, was examined. Estimates of annual average global GPP were higher when JULES was driven with the PRINCETON meteorological dataset than when driven with the WFDEI-GPCC dataset by 3 PgC year⁻¹. On regional scales, differences between the two were observed, with the WFDEI-GPCC-driven model simulations estimating higher GPP in the tropics (5 degrees N-5 degrees S) and the PRINCETON-driven model simulations estimating higher GPP in the extratropics (30-60 degrees N).

93. A global wetland methane emissions and uncertainty dataset for atmospheric chemical transport models (WetCHARTs version 1.0)

Authors: A. Anthony Bloom; Kevin W. Bowman; Meemong Lee; Alexander J. Turner; Ronny Schroeder; John R. Worden; Richard Weidner; Kyle C. McDonald; Daniel J. Jacob

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Abstract: . Wetland emissions remain one of the principal sources of uncertainty in the global atmospheric methane (CH₄) budget, largely due to poorly constrained process controls on CH₄ production in waterlogged soils. Process-based estimates of global wetland CH₄ emissions and their associated uncertainties can provide crucial prior information for model-based top-down CH₄ emission estimates. Here we construct a global wetland CH₄ emission model ensemble for use in atmospheric chemical transport models (WetCHARTs version 1.0). Our 0.5 degrees x 0.5 degrees resolution model ensemble is based on satellite-derived surface water extent and precipitation reanalyses, nine heterotrophic respiration simulations (eight carbon cycle models and a data-constrained terrestrial carbon cycle analysis) and three temperature dependence parameterizations for the period 2009-2010; an extended ensemble subset based solely on precipitation and the data-constrained terrestrial carbon cycle analysis is derived for the period 2001-2015. We incorporate the mean of the full and extended model ensembles into GEOS-Chem and compare the model against surface measurements of atmospheric CH₄; the model performance (site-level and zonal mean anomaly residuals) compares favourably against published wetland CH₄ emissions scenarios. We find that uncertainties in carbon decomposition rates and the wetland extent together account for more than 80 % of the dominant uncertainty in the timing, magnitude and seasonal variability in wetland CH₄ emissions, although uncertainty in the temperature CH₄ : C dependence is a significant contributor to seasonal variations in mid-latitude wetland CH₄ emissions. The combination of satellite, carbon cycle models and temperature dependence parameterizations provides a physically informed structural a priori uncertainty that is critical for top-down estimates of wetland CH₄ fluxes. Specifically, our ensemble can provide enhanced information on the prior CH₄ emission uncertainty and the error covariance structure, as well as a means for using posterior flux estimates and their uncertainties to quantitatively constrain the biogeochemical process controls of global wetland CH₄ emissions.

94. Assimilation of repeated woody biomass observations constrains decadal ecosystem carbon cycle uncertainty in aggrading forests

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Abstract: Forest carbon sink strengths are governed by plant growth, mineralization of dead organic matter, and disturbance. Across landscapes, remote sensing can provide information about aboveground states of forests and this information can be linked to models to estimate carbon cycling in forests close to steady state. For aggrading forests this approach is more challenging and has not been demonstrated. Here we apply a Bayesian approach, linking a simple model to a range of data, to evaluate their information content, for two aggrading forests. We compare high information content analyses using local observations with retrievals using progressively sparser remotely sensed information (repeated, single, and no woody biomass observations). The net biome productivity of both forests is constrained to be a net sink with $<2 \text{ Mg C ha}^{-1} \text{ yr}^{-1}$ variation across the range of inputs. However, the sequestration of particular carbon pool(s) varies with assimilated biomass information. Assimilation of repeated biomass observations reduces uncertainty and/or bias in all ecosystem C pools not just wood, compared to analyses using single or no stock information. As verification, our repeated biomass analysis explains 78-86% of variation in litter dynamics at one forest, while at the second forest total dead organic matter estimates are within observational uncertainty. The uncertainty of retrieved ecosystem traits in the repeated biomass analysis is reduced by up to 50% compared to analyses with less biomass information. This study quantifies the importance of repeated woody observations in constraining the dynamics of both wood and dead organic matter, highlighting the benefit of proposed remote sensing missions.

95. Gridded National Inventory of U.S. Methane Emissions

Authors: Joannes D. Maasakkers (Harvard University); Daniel J. Jacob (Harvard University); Melissa P. Sulprizio (Harvard University); Alexander J. Turner (Harvard University); Melissa Weitz (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Tom Wirth (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Cate Hight (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Mark DeFigueiredo (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Mausami Desai (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Rachel Schmeltz (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Leif Hockstad (ClimateChange Division, Environmental Protection Agency, Washington, District of Columbia 20460, United States); Anthony A. Bloom (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Kevin W. Bowman (National Aeronautics and Space Administration (NASA) Jet Propulsion Laboratory / California Institute of Technology); Seongeun Jeong (Lawrence Berkeley National Laboratory); Marc L. Fischer (Lawrence Berkeley National Laboratory)

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Abstract: N/A

96. Investigating the role of prior and observation error correlations in improving a model forecast of forest carbon balance using Four-dimensional Variational data assimilation

Authors: Ewan M. Pinnington; Eric Casella; Sarah L. Dance; Amos S. Lawless; James I.L. Morison; Nancy K. Nichols; Matthew Wilkinson; Tristan L. Quaife

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Abstract: N/A

97. The decadal state of the terrestrial carbon cycle: Global retrievals of terrestrial carbon allocation, pools, and residence times

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Abstract: Significance Quantitative knowledge of terrestrial carbon pathways and processes is fundamental for understanding the biosphere's response to a changing climate. Carbon allocation, stocks, and residence times together define the dynamic state of the terrestrial carbon cycle. These quantities are difficult to measure and remain poorly quantified on a global scale. Here, we retrieve global 1 degrees x 1 degrees carbon state and process variables by combining a carbon balance model with satellite observations of biomass and leaf area (where and when available) and global soil carbon data. Our results reveal emergent continental-scale patterns and relationships between carbon states and processes. We find that conventional land cover types cannot capture continental-scale variations of retrieved carbon variables; this mismatch has strong implications for terrestrial carbon cycle predictions.